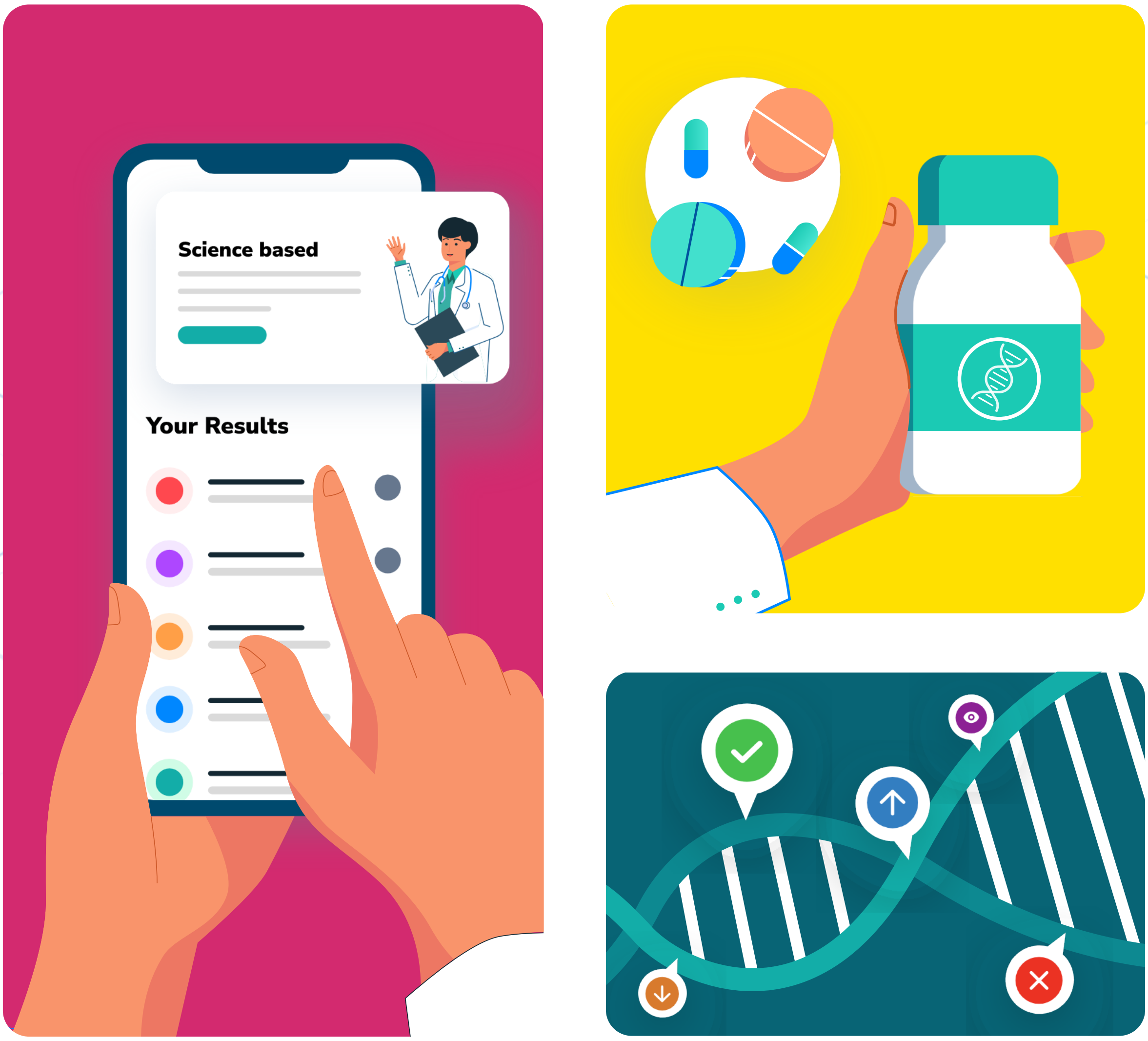
RxReady™



tes UGM3

Date of Birth : **01 Jan 2000**
Report Date : **21 Jan 2025**

NalaGenetics

Table of Contents

Personal Information

Customer ID	**TEUG00012101003**
Name	**Tes UGM3**
Date of birth	**01 Jan 2000**
National ID/ Passport No.	**Not Available**
Gender	**Male**

Ordering Details

Ordered by	**FAD Test Physician One**
License No.	**000123**
Order ID	**#8349-867687**

Sample Details

Clinic Name	**Nala Field Application Department ID**
Address	**Rukan Permata Senayan Blok H1 H2,**
Type of Sample	**Buccal Swab**
Sample ID	**NL2025JAN212609**
Supplementary ID	**-**
Test Method	**Quantitative PCR**

Lab Address	**Jl. Pluit Selatan Raya No.19, Penjaringan, Jakarta Utara, DKI Jakarta**
Collected Date	**21/01/2025 (01:47 PM)**
Received Date	**21/01/2025 (01:48 PM)**
Processing Lab Address	**Jl. Pluit Selatan Raya No.19, Penjaringan, Jakarta Utara, DKI Jakarta**

Introduction

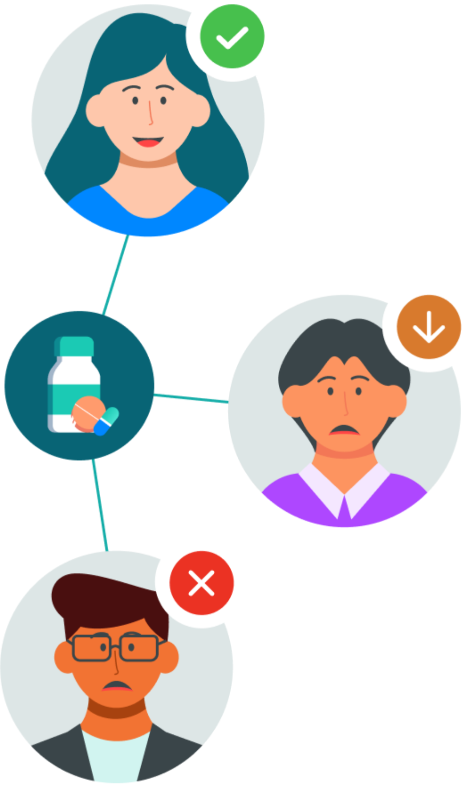
One Size Doesn't Fit All:

The Vital Role of Personalized Medicine

Today's population is living longer than ever before. An aging population means more and more of us are likely to live with long-term health conditions that require medication at some point in our life. As a result, it is crucial that the medications we take are both safe and effective.

Unfortunately, many currently available medicines are adopting 'one size fits all' approach. Throughout history, individuals are given medication at a specific dose for a specific medical condition. However, due to genetic differences, the response to treatments can vary significantly between individuals, with some responding well and others experiencing little to no effect. Some may also experience unpleasant side effects from their prescription.

The failure of the traditional prescription approach does not only burden patients with additional physical discomfort and psychological stress, but also contributes to the rising cost of healthcare. To address these issues, personalized medicine has become increasingly important, with the goal of creating tailored treatments based on an individual's specific genetic makeup and medical history.



What is Pharmacogenomics?

Pharmacogenomics analyzes your unique genetic makeup to identify genetic variations that can affect your response to medication. By proactively testing your DNA, your response to a wide range of medications can be predicted, allowing healthcare providers to make more informed decisions about your treatment plan before you even start taking the medication.

Why Pharmacogenomics Test?

Our test covers predicted responses to an extensive list of medications and consists of:

- Report Overview
- Detailed Medication Report
- Scientific Evidence References
- Continuous Support

By sharing this pharmacogenomic report with your healthcare provider, you can avoid trial-and-error and get on the path to personalized medicine today.

Disclaimer

Recommendations given in this report are made using a lab-developed test which should not supersede clinical judgement or medical expertise.

The DNA test results are not intended to be used by you as a substitute for professional medical advice. You should always seek the advice of a healthcare provider or physician for any diagnostic purpose with any questions you may have regarding the diagnosis, cure, treatment, mitigation or prevention of any disease or other medical conditions or impairment or the status of your health.

The laboratory may not be able to process your sample or the laboratory process may result in errors. The laboratory may not be able to process your DNA sample if it does not contain a sufficient amount of DNA. Furthermore, if the DNA sample that you have provided is contaminated and/or corrupted, the accuracy of the DNA test result may be impacted. Even with stringent acceptance criteria for sample processing that meets our high standards, a small, unknown fraction of the data generated during the laboratory process may be un-interpretable or incorrect and may therefore not be able to generate a corresponding report.

The test detects four genes and twenty variants based on the recommendations of the various medical societies and regulatory bodies. The following genetic variants will be detected in the assay: CYP2D6 rs1065852, rs5030655, rs3892097, rs35742686, rs16947, rs28371725, rs1135840, rs769258, rs5030865, rs5030656, rs59421388, rs267608319, exon 9 conversion (*36), deletion (*5) and duplication; CYP2C9 rs1799853, rs1057910; CYP2C19 rs4244285, rs4986893, rs12248560 ; SLCO1B1 rs4149056.

The copy number assays in this test do not differentiate between partial and whole gene deletions and/or duplications. Copy number variants that are more than three cannot be detected with this test and may have to be detected with a different platform. Tandem-hybrid arrangements in CYP2D6 are not distinguished.

Any complex rearrangements or mosaicisms are also not detected in this test. A normal (wild-type) genotype signifies the absence of the targeted alleles and does not indicate the absence of other mutations not covered by the assay. The possibility cannot be ruled out that the indicated genotypes may be present and are below the limits of detection for this assay. The DNA test may not capture scientific findings which are not yet validated or not provided. Like all diagnostic tests, pharmacogenomic tests are one of multiple pieces of information that clinicians should consider when making therapeutic choice for each patient.

II. Legend

A. Summary of Follow Up Actions

Our knowledge base ranks and summarizes clinical recommendations published by expert consortia, regulatory bodies, and other trusted sources for drugs based on genetic information, along with clinical information of the patient. These symbols are a summary of the recommendation text next to it.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Increase Monitoring
Although patient has a varied genetic profile, the intended medication can still be prescribed as recommended, but with more frequent monitoring for quick action to be taken in the event of unwanted reactions.

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Increase Starting Dose
Patient has altered metabolism rate or reduced activation rate for drug indicated. Increased starting dose has shown to help patient into target therapeutic range faster and/or reduce the risk of drug clinical inefficacy.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

B. Recommendation Caveats

These caveats are taken from pharmacogenomics guidelines, which will usually specify the required conditions and situations where the guideline is relevant. The various caveats are grouped below:

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Turnaround Time
The benefit of waiting for a genetic test result may be less than starting the medication without a genetic test result.

Predictive Value
Low positive predictive value. Testing positive for this phenotype does not mean that the patient will develop the adverse effect.

C. Level of Scientific Evidence

Our knowledge base ranks and summarizes the strength of clinical recommendations published by expert consortia, regulatory bodies, and other trusted sources for drugs based on DNA information, along with other clinical information of the patient. These texts are a summary of the level of scientific evidence available for the respective clinical recommendation as presented below.

STRONG RECOMMENDATION
You can follow the recommendation confidently because its effect has been categorized to improve clinical outcome by at least one of the following medical societies or regulatory bodies. Each guideline has different kinds of strength of evidence. We reclassified the following categories from each guideline into Strong Recommendation:
CPIC/CPNDS - Strong Recommendation OR DPWG - Level 3 or 4 level of evidence OR PRO - strong, essential or advisable test OR FDA/Other regulatory bodies - Testing required

MODERATE RECOMMENDATION
You can consider the recommendation as it has been categorized to have some potential clinical benefit by at least one of the following medical societies or regulatory bodies. We reclassified the following categories from each guideline into Moderate Recommendation:
CPIC/CPNDS - Moderate or optional recommendation OR DPWG - Level 2 level of evidence OR PRO - conditional or possibly helpful test OR FDA/Other regulatory bodies - Testing recommended or actionable PGx

INFORMATION AVAILABLE
The information provided serves as an additional point of consideration for the patient’s therapy. The recommendation with insufficient data is also included as this category. We reclassified the following categories from each guideline into Information Available:
DPWG - Level 0 or 1 level of evidence OR FDA/Other regulatory bodies - Informative PGx

PGx Medical Societies
CPIC (Clinical Pharmacogenetics Implementation Consortium) An international consortium of professionals interested in applying pharmacogenetics for patient care. Established in 2009, it consists of PGRN members, PharmGKB staff, and various experts. The guidelines are indexed in PubMed as clinical guidelines, endorsed by ASHP and ASCPT , and referenced in ClinGen and PharmGKB.
DPWG (Dutch Pharmacogenetics Working Group) A multidisciplinary group of physicians, pharmacists, and other healthcare professionals. In was established in 2005 by the Royal Dutch Pharmacists Association (KNMP) with the objective of developing pharmacogenomic-based therapeutic (dose) recommendations. The guidelines are endorsed by the EACPT and the EAHP.
CPNDS (Canadian Pharmacogenomics Network for Drug Safety) A pan-Canadian active surveillance network consisting of trained surveillance clinicians in 10 pediatric teaching hospitals across Canada, serving >75% of Canada's children. Established in 2005, the network's goal is to improve the safe use of medication by identifying genomic biomarkers of drug risk for serious ADRs.
PRO (Professional Societies) A source that includes the French National Network of Pharmacogenetics (RNPGx), the Cystic Fibrosis Foundation and the American College of Rheumatology for approriate drug presented.
Regulatory Bodies
FDA (Food and Drug Administration) An agency within the US Department of Health and Human Services. It is responsible for protecting the public health by ensuring the safety, efficacy, and security of human and veterinary drugs, biological products, and medical devices.
EMA (European Medicines Agency) A decentralised agency of the European Union (EU) responsible for the scientific evaluation, supervision and safety monitoring of medicines in the EU. EMA is a networking organisation whose activities involve thousands of experts from across Europe. These experts carry out the work of EMA's scientific committees.
PMDA (Pharmaceuticals and Medical Devices Agency) A Japan agency that was established and came into service on April 1, 2004, under the Law for the Pharmaceuticals and Medical Devices Agency, as a part of the Japan Association for the Advancement of Medical Equipment (JAAME).
Swissmedic The Swiss authority responsible for the authorisation and supervision of therapeutic products. The activities of Swissmedic are based on the Law on Therapeutic Products.
HCSC (Health Canada Santé Canada) A federal department responsible for helping Canadians maintain and improve their health, while respecting individual choices and circumstances.

Each guideline and label annotation has its own strength of evidence for every drug-gene pair interaction.

D. Localization of Evidence

- The drugs presented in this report are all drugs that are available in your country.
- Our PGx recommendations are always supported by research findings from studies done in your ethnic population (e.g. if you are Chinese, we provide supporting scientific evidence from studies in the Chinese population).
- RxReady tests for genetic polymorphisms that are relevant in the Asian population.

III. Executive Report

PATIENT NAME	NATIONAL ID / PASSPORT NO.	ORDERING PROVIDER
tes UGM3	**Not available**	**FAD Test Physician One**
GENDER	DATE OF BIRTH	ORDER ID
Male	**01 Jan 2000**	**8349-867687**

A. Genomic Information

Gene	Genotype	Phenotype
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer
SLCO1B1	Rs4149056(TT)	SLCO1B1 Normal Function

B. Current Medications

Information not available.

Simvastatin 20 mg once daily at night
Atenolol 50 mg once daily at night
Propafenone 150 mg three times daily
Omeprazole 20 mg once daily before breakfast
Paroxetine 20 mg once daily at night
Phenytoin 100 mg three times daily
Sumatriptan 50 mg as needed for migraine (max 200 mg/day)
Amoxicillin/ Clavulanate 875 mg/125 mg twice daily for 7 days

C. Pharmacogenomics Summary

1. Cardiovascular System

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Angiotensin II Receptor Blockers					• Losartan || CYP2C9
Antiangina					• Ranolazine
Antiarrhythmics Class I	• Propafenone		• Flecainide		• Disopyramide • Quinidine
Antiarrhythmics Class I/III					• Vernakalant
Antiarrhythmics Class III					• Amiodarone • Dronedarone || CYP2D6
Anticoagulants					• Acenocoumarol || CYP2C9 • Phenprocoumon || CYP2C9 • Warfarin
Antihemorrhagics					• Avatrombopag
Antiplatelets					• Aspirin || CYP2C9 • Clopidogrel • Prasugrel || CYP2C19 • Prasugrel || CYP2C9 • Ticagrelor
Beta Blocking Agents					• Atenolol • Bisoprolol • Carvedilol • Metoprolol • Nebivolol • Propranolol • Sotalol • Timolol
Central Agonists					• Clonidine

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Fibrates					• Fenofibrate And Simvastatin
Nonselective NSAIDs					• Aspirin || CYP2C9
Statins					• Atorvastatin • Fluvastatin • Fluvastatin || CYP2C9 • Fluvastatin || SLCO1B1 • Lovastatin • Pitavastatin • Pravastatin • Rosuvastatin || SLCO1B1 • Simvastatin

2. Dermatology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Others					• Abrocitinib || CYP2C9 • Abrocitinib || CYP2C19

3. Endocrinology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Blood Glucose Lowering Drugs					• Nateglinide
Sulfonylureas					• Glibenclamide || CYP2C9 • Gliclazide || CYP2C9 • Glimepiride || CYP2C9 • Tolbutamide || CYP2C9

4. Family Medicine

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
1st Gen Antipsychotics			• Zuclopenthixol		• Flupenthixol • Haloperidol • Perphenazine
2nd Gen Antipsychotics					• Aripiprazole • Aripiprazole Lauroxil • Brexpiprazole • Cariprazine • Clozapine || CYP2D6 • Olanzapine || CYP2D6 • Paliperidone • Quetiapine || CYP2D6 • Risperidone
Alpha Blockers					• Tamsulosin
Analgesics					• Rimegepant
Angiotensin II Receptor Blockers					• Losartan || CYP2C9
Antiangina					• Ranolazine
Antiarrhythmics Class I	• Propafenone		• Flecainide		
Antiarrhythmics Class III					• Amiodarone • Dronedarone || CYP2D6
Anticholinergics					• Tiotropium • Umeclidinium
Anticoagulants					• Warfarin
Antiemetics And Antinauseants					• Ondansetron • Palonosetron
Antiepileptics					• Clobazam • Diazepam
Antifungals	• Voriconazole || CYP2C19				• Terbinafine • Voriconazole || CYP2C9

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Antimetabolites					• Methotrexate
Antimuscarinics					• Tolterodine
Antiplatelets					• Aspirin || CYP2C9 • Clopidogrel • Prasugrel || CYP2C19 • Prasugrel || CYP2C9 • Ticagrelor
Anxiolytics					• Diazepam
Atypical Antidepressants					• Mirtazapine || CYP2C19 • Mirtazapine || CYP2D6
Beta Blocking Agents					• Atenolol • Bisoprolol • Carvedilol • Metoprolol • Nebivolol • Propranolol • Sotalol • Timolol
COX-2 Selective NSAIDs					• Celecoxib
Cough Suppressants					• Dextromethorphan
Fibrates					• Fenofibrate And Simvastatin
Long-Acting Beta Agonists					• Formoterol || CYP2C19 • Formoterol || CYP2D6
Monoamine Oxidase A Inhibitors					• Moclobemide

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Nonselective NSAIDs					• Aceclofenac • Aspirin || CYP2C9 • Diclofenac || CYP2C9 • Flurbiprofen • Ibuprofen • Indomethacin • Meloxicam • Naproxen • Piroxicam
Opioid Cough Suppressants					• Codeine
Opioids					• Oxycodone • Tramadol || CYP2D6
Other Antidepressants					• Vortioxetine
Others					• Abrocitinib || CYP2C9 • Drospirenone And Ethinylestradiol • Eletriptan • Abrocitinib || CYP2C19
Propulsives					• Metoclopramide || CYP2D6
Proton Pump Inhibitors		• Dexlansoprazole • Lansoprazole • Omeprazole • Pantoprazole			• Esomeprazole • Rabeprazole
Psychostimulants					• Atomoxetine

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Selective Serotonin Reuptake Inhibitors	• Citalopram || CYP2C19 • Escitalopram || CYP2C19		• Paroxetine		• Citalopram || CYP2D6 • Escitalopram || CYP2D6 • Fluoxetine • Fluvoxamine || CYP2C19 • Fluvoxamine || CYP2D6 • Sertraline || CYP2C19 • Sertraline || CYP2D6
Selective Serotonin And Norepinephrine Reuptake Inhibitors					• Venlafaxine
Serotonin/Norepinephrine Reuptake Inhibitors					• Desvenlafaxine • Duloxetine
Statins					• Atorvastatin • Fluvastatin • Fluvastatin || CYP2C9 • Fluvastatin || SLCO1B1 • Lovastatin • Pitavastatin • Pravastatin • Rosuvastatin || SLCO1B1

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Sulfonylureas					• Glibenclamide || CYP2C9 • Gliclazide || CYP2C9 • Glimepiride || CYP2C9 • Tolbutamide || CYP2C9
Tricyclic Antidepressants	• Amitriptyline • Amitriptyline || CYP2C19 • Clomipramine • Clomipramine || CYP2C19 • Imipramine • Imipramine || CYP2C19		• Amitriptyline || CYP2D6 • Clomipramine || CYP2D6 • Imipramine || CYP2D6 • Nortriptyline		

5. Gastrointestinal System

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Antiemetics And Antinauseants					• Dronabinol • Ondansetron • Palonosetron • Tropisetron
Others					• Eliglustat
Propulsives					• Metoclopramide || CYP2D6
Proton Pump Inhibitors		• Dexlansoprazole • Lansoprazole • Omeprazole • Pantoprazole			• Esomeprazole • Rabeprazole

6. Gynecology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Alpha Blockers					• Tamsulosin
Antimuscarinics					• Darifenacin • Fesoterodine • Tolterodine
Beta Agonist					• Mirabegron
Hypothalamic Hormones					• Elagolix
Others					• Drospirenone And Ethinylestradiol • Flibanserin || CYP2C19 • Flibanserin || CYP2C9 • Flibanserin || CYP2D6
Selective Estrogen Receptor Modulators					• Ospemifene || CYP2C9

7. Immunology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Immunosuppressants					• Siponimod

8. Infectious Disease

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Antifungals	• Voriconazole || CYP2C19				• Terbinafine • Voriconazole || CYP2C9
Antimalarials					• Quinine || CYP2D6
Antiretrovirals					• Atazanavir • Nelfinavir • Ritonavir
Antivirals					• Letermovir

9. Musculoskeletal System

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Antigouts					• Lesinurad
Muscle Relaxants					• Tolperisone

10. Neurology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Antidementia Drugs					• Donepezil • Galantamine
Antiepileptics					• Brivaracetam || CYP2C19 • Brivaracetam || CYP2C9 • Clobazam • Diazepam • Fosphenytoin • Lacosamide • Phenytoin || CYP2C19 • Phenytoin || CYP2C9
Anxiolytics					• Diazepam
Others				• Tetrabenazine	• Cevimeline • Deutetrabenazine • Eletriptan • Valbenazine

11. Oncology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Antimetabolites					• Methotrexate
Estrogen Receptor Antagonists	• Tamoxifen				
Immunosuppressants					• Upadacitinib
Other Antineoplastic Agents					• Rucaparib
Protein Kinase Inhibitors					• Axitinib • Erdafitinib • Gefitinib • Ibrutinib

12. Ophthalmology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Beta Blocking Agents					• Timolol

13. Pain Management

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Analgesics					• Paracetamol, Caffeine, And Dihydrocodeine • Paracetamol, Combinations Excl. Psycholeptics • Rimegepant
Antiplatelets					• Aspirin || CYP2C9
COX-2 Selective NSAIDs					• Celecoxib • Lumiracoxib
Muscle Relaxants					• Carisoprodol
Nonselective NSAIDs					• Aceclofenac • Aspirin || CYP2C9 • Diclofenac || CYP2C9 • Dipyrone • Flurbiprofen • Ibuprofen • Indomethacin • Lornoxicam • Meloxicam • Nabumetone • Naproxen • Piroxicam • Tenoxicam
Opioids					• Oliceridine • Oxycodone • Tramadol || CYP2D6

14. Psychiatry

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
1st Gen Antipsychotics			• Pimozide • Zuclopenthixol		• Flupenthixol • Haloperidol • Perphenazine • Thioridazine
2nd Gen Antipsychotics					• Aripiprazole • Aripiprazole Lauroxil • Brexpiprazole • Cariprazine • Clozapine || CYP2D6 • Iloperidone • Olanzapine || CYP2D6 • Paliperidone • Quetiapine || CYP2D6 • Risperidone • Sertindole
Antiepileptics					• Diazepam
Anxiolytics					• Diazepam
Atypical Antidepressants				• Bupropion	• Mirtazapine || CYP2C19 • Mirtazapine || CYP2D6
Drugs Used In Addictive Disorders					• Lofexidine
Drugs Used In Opioid Dependence					• Methadone || CYP2D6
Monoamine Oxidase A Inhibitors					• Moclobemide
Other Antidepressants					• Vortioxetine

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Psychostimulants					• Amfetamine • Atomoxetine • Methylphenidate • Modafinil • Pitolisant
Selective Serotonin Reuptake Inhibitors	• Citalopram || CYP2C19 • Escitalopram || CYP2C19		• Paroxetine		• Citalopram || CYP2D6 • Escitalopram || CYP2D6 • Fluoxetine • Fluvoxamine || CYP2C19 • Fluvoxamine || CYP2D6 • Sertraline || CYP2C19 • Sertraline || CYP2D6
Selective Serotonin And Norepinephrine Reuptake Inhibitors					• Venlafaxine
Serotonin Reuptake Inhibitors					• Nefazodone
Serotonin/Norepinephrine Reuptake Inhibitors					• Desvenlafaxine • Duloxetine

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Tricyclic Antidepressants	• Amitriptyline • Amitriptyline || CYP2C19 • Clomipramine • Clomipramine || CYP2C19 • Doxepin • Doxepin || CYP2C19 • Imipramine • Imipramine || CYP2C19 • Trimipramine • Trimipramine || CYP2C19		• Amitriptyline || CYP2D6 • Clomipramine || CYP2D6 • Desipramine • Doxepin || CYP2D6 • Imipramine || CYP2D6 • Nortriptyline • Trimipramine || CYP2D6		• Amoxapine • Protriptyline

15. Respiratory System

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Anticholinergics					• Tiotropium • Umeclidinium
Antihistamines				• Meclizine	
Cough Suppressants					• Dextromethorphan • Dextromethorphan And Quinidine
Long-Acting Beta Agonists					• Arformoterol • Formoterol || CYP2C19 • Formoterol || CYP2D6
Opioid Cough Suppressants					• Codeine • Hydrocodone

16. Urology

Drug Class	Change Prescription	Increase Starting Dose	Decrease Starting Dose	Monitor	Follow Standard Dosing
Alpha Blockers					• Tamsulosin
Antimuscarinics					• Darifenacin • Fesoterodine • Tolterodine
Beta Agonist					• Mirabegron
Selective Serotonin Reuptake Inhibitors					• Dapoxetine

IV. Individual Drug Report

Abrocitinib || CYP2C19

Abrocitinib

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Body weight, gender, CYP2C19 genotype, race and age did not have a clinically meaningful effect on abrocitinib exposure.

Recommendations based on specific guidelines

EMA	**No actionable recommendation available for this drug-gene pair.**

More information about this drug-phenotype can also be found at **HCSC**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Abrocitinib || CYP2C9

Abrocitinib

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Body weight, gender, CYP2C9 genotype, race and age did not have a clinically meaningful effect on abrocitinib exposure.

Recommendations based on specific guidelines

EMA	**No actionable recommendation available for this drug-gene pair.**

More information about this drug-phenotype can also be found at **HCSC**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Aceclofenac

ACB, Aceclofenac, Airtal

START INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	No actionable recommendation available for this drug-gene pair.

Caveat

Rare CYP2C9 variants may not be detected in genotype tests, classifying patients with these variants as having a normal metabolizer (NM) phenotype based on CYP2C9*1/1 results. The CYP2C9*1 allele may harbor reduced or non-functional variants, so reports should specify the variants or SNPs analyzed. CYP2C9 genotype is one of many factors to consider when prescribing NSAIDs. Age, sex, ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI risk. Use with caution in elderly patients, as they have reduced CYP2C9 metabolism and increased renal/GI risks. Combining NSAIDs with antiplatelets or anticoagulants raises bleeding risk or can interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Acenocoumarol || CYP2C9

Acenocoumarol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Amfetamine

Amphetamine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Although the enzymes involved in amphetamine metabolism have not been clearly defined, CYP2D6 is known to be involved with formation of 4-hydroxy-amphetamine. Since CYP2D6 is genetically polymorphic, population variations in amphetamine metabolism are a possibility.

Recommendations based on specific guidelines

FDA	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Amiodarone

Amiodarone, Aratac, Cordarone, Kendaron, Lamda, Rexidron, Tiaryt

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Amitriptyline

Amitriptyline, Apo-Amitryptiline, Trilin, Tripta

START **MODERATE RECOMMENDATION**

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider alternative drug not metabolized by CYP2C19.

CYP2D6 IM: Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds. Higher plasma concentrations of active drug may increase probability of side effects. CYP2C19 UM: Increased metabolism of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Consider alternative drug not metabolized by CYP2C19. If amitriptyline is warranted, utilize therapeutic drug monitoring to guide dose adjustment.**

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Amitriptyline || CYP2C19

Amitriptyline, Apo-Amitryptiline, Trilin, Tripta

START MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19.

Increased metabolism of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19. Tricyclic antidepressants (TCAs) without major CYP2C19 metabolism include the secondary amines nortriptyline and desipramine. If a tertiary amine is warranted, utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	No actionable recommendation available for this drug-gene pair.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Amitriptyline || CYP2D6

Amitriptyline, Apo-Amitryptiline, Trilin, Tripta

START

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider 25% reduction of recommended starting dose.

Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds when compared to normal metabolizers. Higher plasma concentrations of active drug will increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Consider 25% reduction of recommended starting dose. Patients may receive an initial low dose of a tricyclic, which is then increased over several days to the recommended steady-state dose. Utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	Use 75% of the standard dose and monitor the efficacy and side effects or the plasma concentrations.
FDA	No actionable recommendation available for this drug-gene pair.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Amoxapine
Amoxapine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.
May alter systemic concentrations.

Recommendations based on specific guidelines

FDA	No actionable recommendation available for this drug-gene pair.

Source
Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Arformoterol

Arformoterol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Aripiprazole

Abilify, Arinia, Aripi, Aripiprazole, Ariski, Avram, Zipren, Zonia

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The genetic variation increases the plasma concentration of the sum of aripiprazole and the active metabolite dehydroaripiprazole to a limited degree. There is insufficient evidence that this increases the risk of side effects.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

30565279 24682161

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Aripiprazole lauroxil

Aripiprazole Lauroxil

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Aspirin || CYP2C9

Thrombo Aspilets, Aptor, Ascardia, Aspilets, Aspirin, Astika, Bayer Aspirin, Bodrexin, Cardio Aspirin, Cartylo, Contrexyn, Farmasal, Gramasal, Inzana, Miniaspi, Naspro, Nospirinal

START INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	No actionable recommendation available for this drug-gene pair.

Caveat

Rare CYP2C9 variants may not be detected in genotype tests, classifying patients with these variants as having a normal metabolizer (NM) phenotype based on CYP2C9*1/1 results. The CYP2C9*1 allele may harbor reduced or non-functional variants, so reports should specify the variants or SNPs analyzed. CYP2C9 genotype is one of many factors to consider when prescribing NSAIDs. Age, sex, ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI risk. Use with caution in elderly patients, as they have reduced CYP2C9 metabolism and increased renal/GI risks. Combining NSAIDs with antiplatelets or anticoagulants raises bleeding risk or can interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Atazanavir

Atazanavir, Reyataz

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Atenolol

Aiti, Alonet, Apo-Atenol, Atenolol, Atenosan, Betablok, Catenol, Farnormin, Hypernol, Internolol, Lotenal, Lotensi, Nif-Ten, Normaten, Noten, Prenolol, Pretenol, Tenol, Tenolol, Tenormin, Tensig, Urosin, Vascoten, Velorin

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Atomoxetine

Atomoxetine, Strattera

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer (non-*10)

Recommendation

Initiate with 40 mg/day and increase to 80 mg/day after 3 days.

Possibly higher atomoxetine concentrations but questionable clinical significance.

Recommendations based on specific guidelines

CPIC	**Initiate with 40 mg/day and increase to 80 mg/day after 3 days. If no clinical response and in the absence of adverse events after 2 weeks, consider increasing to 100 mg/day. If no clinical response observed after 2 weeks, consider obtaining a peak plasma concentration 1 to 2 hours after dose administered. If <200 ng/ml, consider a proportional increase to approach 400 ng/ml. Therapeutic range of 200 to 1000 ng/ml has been proposed. Modest improvement in response (reduction in ADHD-RS) is observed at peak concentrations >400 ng/ml. Dosages >100 mg/day may be needed to achieve target concentrations.**
DPWG	Occurrence of side effects and/or a response later than 9 weeks: reduce dose. Check whether the effect is conserved. Plasma concentration is 2-3 times higher for IM than for EM at the same dose.

Caveat

Available exposure-response data from clinical trials are derived from concentrations drawn 60–90 minutes after dosing; although this measure may not be the best predictor of response, the relationship between other exposure measures, such as trough concentration at steady state, steady-state AUC, unbound atomoxetine concentrations, or total active compounds (unbound atomoxetine + unbound 4-OH aglycone), has not been evaluated.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Atorvastatin

Actalipid, Apo-Atorvastatin, Atavor, Atofit, Atoris, Atorsan, Atorvachol, Atorvastatin, Atorvon, Atswift, Beatorva, Eturion, Fastor, Genlipid, Lipitor, Litorcom, Removchol, Simtor, Stator, Stavinor, Tavora, Torvalip, Torvatec, Tulip

START | STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.
Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

32128760 33608664

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Avatrombopag

Avatrombopag

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Axitinib

Axitinib, Inlyta

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Population pharmacokinetic analyses in patients with advanced cancer (including advanced RCC) and healthy volunteers indicate that there are no clinically relevant effects of age, gender, body weight, race, renal function, UGT1A1 genotype, or CYP2C19 genotype.

Recommendations based on specific guidelines

EMA	**No actionable recommendation available for this drug-gene pair.**

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

29524031

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Bisoprolol

B-Beta, Beta-One, Biofin, Bisocor, Bisohexal, Bisopro, Bisoprolol, Bisovell, Carbisol, Concor, Hapsen, Konblobet, Maintate, Selbix

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Brexpiprazole

Brexpiprazole, Rexulti

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

There are indications supporting an increase in the exposure to brexpiprazole, but no indications supporting an increase in side effects in patients with this gene variation.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**
SWISSMEDIC	Known slow CYP2D6 metabolizers: administration of half the usual dose. Known slow CYP2D6 metabolizers taking moderate/strong CYP3A4 inhibitors: administration of a quarter of the usual dose.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

28750151

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Brivaracetam || CYP2C19

Brivaracetam

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

24717838

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Brivaracetam || CYP2C9

Brivaracetam

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Bupropion

Bupropion, Wellbutrin, Zyban

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

It may be necessary to decrease the dose of CYP2D6 substrates when used concomitantly with bupropion.
Co-administration of bupropion with drugs that are metabolized by CYP2D6 can increase the exposures of drugs that are substrates of CYP2D6.

Recommendations based on specific guidelines

FDA	**It may be necessary to decrease the dose of CYP2D6 substrates when used concomitantly with bupropion. Particularly for drugs with a narrow therapeutic index. Patients treated concomitantly with bupropion and drugs that require metabolic activation by CYP2D6 to be effective may require increased doses of the drug.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Increase Monitoring
Although patient has a varied genetic profile, the intended medication can still be prescribed as recommended, but with more frequent monitoring for quick action to be taken in the event of unwanted reactions.

Cariprazine

Cariprazine, Symvenu

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Carisoprodol

Carisoprodol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Carvedilol

Blorec, Bloved, Carvedilol, Carvedilol Sandoz, Carvepen, Carvilol, Dilatrend, Vacodil, V-Bloc

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The plasma concentration of carvedilol can be elevated. This does not, however, result in an increase in side effects.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Celecoxib

Actrel, Celcox, Celdol, Celebrex, Celecoxib, Celecoxib Sandoz, Novexib, Remabrex

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Initiate therapy with recommended starting dose.
Normal metabolism.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.**

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

37062721

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Cevimeline

Cevimeline

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Citalopram || CYP2C19

Citalopram

START

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Consider a clinically appropriate alternative antidepressant not predominantly metabolized by CYP2C19.
Increased metabolism of citalopram and escitalopram to less active compounds when compared to CYP2C19 rapid and normal metabolizers. Lower plasma concentrations decrease the probability of clinical benefit.

Recommendations based on specific guidelines

CPIC	**Consider a clinically appropriate alternative antidepressant not predominantly metabolized by CYP2C19. If citalopram or escitalopram are clinically appropriate, and adequate efficacy is not achieved at standard maintenance dosing, consider titrating to a higher maintenance dose. Drug-drug interactions and other patient characteristics (e.g., age, renal function, liver function) should be considered when adjusting dose or selecting an alternative therapy.**
DPWG	No actionable recommendation available for this drug-gene pair.

Caveat

Patients on a stable and effective dose of an SSRI most likely will not benefit from additional dose modifications based on CYP2D6 or CYP2C19 genotype results. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Citalopram || CYP2D6

Citalopram

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**
SWISSMEDIC	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Clobazam

Anxibloc, Asabium, Clobazam, Clofritis, Frisium, Proclozam

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Clomipramine

Anafranil, Apo-Clomipramine, Clomipramine, Depranil

START MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider alternative drug not metabolized by CYP2C19.

CYP2D6 IM: Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds. Higher plasma concentrations of active drug may increase probability of side effects. CYP2C19 UM: Increased metabolism of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Consider alternative drug not metabolized by CYP2C19. If tricyclic antidepressants (TCAs) is warranted, utilize therapeutic drug monitoring to guide dose adjustment.**

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Clomipramine || CYP2C19

Anafranil, Apo-Clomipramine, Clomipramine, Depranil

START **MODERATE RECOMMENDATION**

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19.

Increased metabolism of tertiary amines compared to normal metabolizers. Greater conversion of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19. Tricyclic antidepressants (TCAs) without major CYP2C19 metabolism include the secondary amines nortriptyline and desipramine. If a tertiary amine is warranted, utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	Indication OBSESSIVE COMPULSIVE DISORDER or ANXIETY DISORDERS: avoid clomipramine. Indication DEPRESSION: no action required. If clomipramine is warranted, monitor the effect and side effects or the plasma concentrations.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Clomipramine || CYP2D6

Anafranil, Apo-Clomipramine, Clomipramine, Depranil

START | MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider 25% reduction of recommended starting dose.

Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds when compared to normal metabolizers. Higher plasma concentrations of active drug will increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Consider 25% reduction of recommended starting dose. Patients may receive an initial low dose of a tricyclic, which is then increase over several days to the recommended steady-state dose. Utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	Use 70% of the standard dose. Monitor the effect and side effects of the plasma concentrations of clomipramine and desmethylclomipramine.
FDA	Monitor TCA plasma levels when co-administering Clomipramine with a P450 2D6 inhibitor.

More information about this drug-phenotype can also be found at **SWISSMEDIC**.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Clonidine

Catapres, Clonidine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Clopidogrel

Apo-Clopidogrel, Artepid, Caplor, Ceruvin, Clopido, Clopidogrel, Clopidogrel Sandoz, Clopidogrel Stada, Clopigrel, Clopivid, Clotromboz, Cpg, Deplatt, G Plus - Clopidogrel, Gridokline, Medigrel, Pidovix, Placta, Pladogrel, Platless, Plavesco, Plavix, Therodel, Thinoclo, Trombikaf

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

If considering clopidogrel, use at standard dose (75 mg/day).

Increased clopidogrel active metabolite formation; lower on-treatment platelet reactivity; no association with higher bleeding risk.

Recommendations based on specific guidelines

CPIC	**If considering clopidogrel, use at standard dose (75 mg/day).**
DPWG	No actionable recommendation available for this drug-gene pair.
PRO	There is no guideline concerning the appropriate dosage to prescribe for a patient with the *17 allele.
EMA	No actionable recommendation available for this drug-gene pair.

Caveat

An assigned CYP2C19*1 allele could potentially harbor an undetected genetic variant. Most clinical laboratories do not currently test for CYP2C19 deletions or structural variants. Therefore, clinical providers should understand which CYP2C19 variant alleles were genotyped when interpreting results. CYP2C19 genotype is just one factor that clinicians should consider when prescribing clopidogrel.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

33523336 34708996 37279000

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Clozapine || CYP2D6

Clopine, Clorilex, Clozapine, Clozapine Sandoz, Clozaril, Clozer, Cycozam, Lozap, Nucloz, Nuzip, Sizoril

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No action is required for this gene-drug interaction.

Variation in CYP2D6 can affect the plasma concentration of clozapine, there are no clinical consequences to this change.

Recommendations based on specific guidelines

DPWG	No action is required for this gene-drug interaction.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Codeine

Codeine, Codikaf, L.T.R. Cough Linctus, Linctus Tussis Rubra, Sp-Codin Linctus

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Use codeine label recommended age- or weight-specific dosing.

Reduced morphine formation.

Recommendations based on specific guidelines

CPIC	**Use codeine label recommended age- or weight-specific dosing. If no response and opioid use is warranted, consider a non-tramadol opioid.**
DPWG	For COUGH: No action required. For PAIN: Be alert to a reduced effectiveness. In the case of inadequate effectiveness: try a dose increase, if this does not work: choose an alternative. Do not select tramadol, as this is also metabolized by CYP2D6. Morphine is not metabolized by CYP2D6. Oxycodone is metabolized by CYP2D6 to a limited extent, but this does not result in differences in analgesia in patients. If no alternative is selected: advise the patient to report inadequate analgesia. It is not possible to offer adequately substantiated advice for dose adjustment based on the limited available literature for this phenotype.

More information about this drug-phenotype can also be found at **CPNDS**.

Caveat

Like all diagnostic tests, CYP2D6 genotype is one of multiple pieces of information that clinicians should consider in guiding their therapeutic choice for each patient. Furthermore, there are several other factors that cause potential uncertainty in the genotyping results and phenotype predictions.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Dapoxetine

Dapoxetine, Priligy

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Darifenacin

Darifenacin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Desipramine

Desipramine

START

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider 25% reduction of recommended starting dose.

Reduced metabolism of TCAs to less active compounds when compared to normal metabolizers. Higher plasma concentrations of active drug will increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Consider 25% reduction of recommended starting dose. Patients may receive an initial low dose of a tricyclic, which is then increase over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose. Utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
FDA	Desirable to monitor TCA plasma levels when co-administering a TCA with a known CYP2D6 inhibitor drug.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose

Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Initiation

This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Desvenlafaxine

Desvenlafaxine, Pristiq

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Deutetrabenazine

Deutetrabenazine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Dexlansoprazole

Dexilant, Dexlansoprazole

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.

Decreased plasma concentrations of PPIs compared with CYP2C19 NMs; increase risk of therapeutic failure.

Recommendations based on specific guidelines

CPIC	**Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.**

Caveat

An assigned *1 allele could potentially harbor an undetected CYP2C19 genetic variant that results in altered metabolism and drug exposure. Most clinical laboratories do not currently test for CYP2C19 copy number variants or deletions. Therefore, clinical providers should understand which CYP2C19 variant alleles were genotyped when interpreting results. CYP2C19 genotype is just one factor that clinicians should consider when prescribing PPIs.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Increase Starting Dose
Patient has altered metabolism rate or reduced activation rate for drug indicated. Increased starting dose has shown to help patient into target therapeutic range faster and/or reduce the risk of drug clinical inefficacy.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Dextromethorphan

Axcel Dextromethorphan, Beathorphan, Dexcophan, Dextromethorphan, Dextromethorphan Linctus, Metophan, Nospan, Pusiran, Sunthorphan, Tussidex Forte Linctus, Tussils

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Dextromethorphan and Quinidine

Dextromethorphan and Quinidine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Diazepam

Anxicalm, Apo-Diazepam, Dbl Diazepam, Diapo, Diazepam, Diazepam Lipuro, Dzp, Nozepav, Stesolid, Trazep, Valdimex, Valisanbe, Valium

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Diclofenac || CYP2C9

Anuva, Diclofenac, Flamar, Zorvolex

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	No actionable recommendation available for this drug-gene pair.

Caveat

Rare CYP2C9 variants may not be detected in genotype tests, classifying patients with these variants as having a normal metabolizer (NM) phenotype based on CYP2C9*1/1 results. The CYP2C9*1 allele may harbor reduced or non-functional variants, so reports should specify the variants or SNPs analyzed. CYP2C9 genotype is one of many factors to consider when prescribing NSAIDs. Age, sex, ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI risk. Use with caution in elderly patients, as they have reduced CYP2C9 metabolism and increased renal/GI risks. Combining NSAIDs with antiplatelets or anticoagulants raises bleeding risk or can interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

37062721

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Dipyrone

Dipyrone

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	No actionable recommendation available for this drug-gene pair.

Caveat

Rare CYP2C9 variants may not be detected in genotype tests, classifying patients with these variants as having a normal metabolizer (NM) phenotype based on CYP2C9*1/1 results. The CYP2C9*1 allele may harbor reduced or non-functional variants, so reports should specify the variants or SNPs analyzed. CYP2C9 genotype is one of many factors to consider when prescribing NSAIDs. Age, sex, ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI risk. Use with caution in elderly patients, as they have reduced CYP2C9 metabolism and increased renal/GI risks. Combining NSAIDs with antiplatelets or anticoagulants raises bleeding risk or can interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Disopyramide

Disopyramide

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Donepezil

Alzim, Aricept, Aricept Evess, Aripezil, Donacept, Doncept, Donepezil, Donepezil Mevon, Dopezil, Fepezil, Fordesia, Hetero Donepezil Hydrochloride, Torpezil

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

31064198

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Doxepin

Apo-Doxepin, Doxepin, Sagalon

START MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider alternative drug not metabolized by CYP2C19.

CYP2D6 IM: Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds. Higher plasma concentrations of active drug may increase probability of side effects. CYP2C19 UM: Increased metabolism of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Consider alternative drug not metabolized by CYP2C19. If tricyclic antidepressants (TCAs) is warranted, utilize therapeutic drug monitoring to guide dose adjustment.**

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Doxepin || CYP2C19

Apo-Doxepin, Doxepin, Sagalon

START

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19.

Increased metabolism of tertiary amines compared to normal metabolizers. Greater conversion of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19. Tricyclic antidepressants (TCAs) without major CYP2C19 metabolism include the secondary amines nortriptyline and desipramine. If a tertiary amine is warranted, utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	No actionable recommendation available for this drug-gene pair.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Doxepin || CYP2D6

Apo-Doxepin, Doxepin, Sagalon

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider 25% reduction of recommended starting dose.

Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds when compared to normal metabolizers. Higher plasma concentrations of active drug will increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Consider 25% reduction of recommended starting dose. Patients may receive an initial low dose of a tricyclic, which is then increase over several days to the recommended steady-state dose. Utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	Use 80% of the standard dose and monitor the effect and side effects or the plasma concentrations of doxepin and nordoxepin in order to set the maintenance dose. The therapeutic range is 100-250 ng/mL for the sum of doxepin and nordoxepin plasma concentrations. Values higher than 400 ng/mL are considered toxic.
FDA	No actionable recommendation available for this drug-gene pair.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Dronabinol

Dronabinol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Dronedarone || CYP2D6

Dronedarone, Multaq

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Drospirenone and Ethinylestradiol

Drospera, Gveza, Jastinda, Liza, Synfonia, Yasmin, Yaz

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In the study with 24 women with homozygous (wild type) and heterozygous CYP2C19 genotype, the daily oral administration of 3 mg for 14 days did not affect the oral clearance of omeprazole (40 mg, single oral dose) and the CYP2C19 product 5-hydroxy omeprazole. These results demonstrate that drospirenone did not inhibit CYP2C19 in vivo.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Duloxetine

Cymbalta, Duloxetine, Dulxota

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Duloxetine is a substrate of CYP1A2 and CYP2D6; however, existing data do not support a clinically meaningful impact of CYP2D6 on duloxetine and thus was assigned CPIC level C (no recommendation).

Recommendations based on specific guidelines

CPIC	**No actionable recommendation available for this drug-gene pair.**
DPWG	No actionable recommendation available for this drug-gene pair.
SWISSMEDIC	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Elagolix
Elagolix

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Follow Standard Dosing Guideline
No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source
Publications related to local relevance for this drug-gene pair have not been published yet.

Eletriptan

Eletriptan, Relpax

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In vitro studies show that eletriptan is mainly metabolized by the hepatic cytochrome P-450 enzyme CYP3A4. In vitro studies also suggest low involvement of CYP2D6, although clinical studies indicate that there is no clinically relevant effect of CYP2D6 polymorphism on the pharmacokinetics of eletriptan.

Recommendations based on specific guidelines

SWISSMEDIC **No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Eliglustat

Eliglustat

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Use the standard dose If NO co-medication with a moderate or strong CYP2D6 or CYP3A inhibitor or strong CYP3A inducer.

This gene variation reduces the conversion of eliglustat to inactive metabolites. However, in the absence of CYP2D6 and CYP3A inhibitors, this does not result in a clinically significant increase risk of side effects.

Recommendations based on specific guidelines

DPWG	**Use the standard dose If NO co-medication with a moderate or strong CYP2D6 or CYP3A inhibitor or strong CYP3A inducer.** **Co-medication with BOTH a MODERATE to STRONG CYP2D6 INHIBITOR AND a MODERATE to STRONG CYP3A INHIBITOR: Eliglustat is contra-indicated.** **1. Choose an alternative if possible.** **Co-medication with a STRONG CYP2D6 INHIBITOR:** **1. Use a dose of 84mg eliglustat 1x daily.** **Co-medication with a MODERATE CYP2D6 INHIBITOR:** **1. Consider a dose of 84mg eliglustat 1x daily.** **2. Be alert to side effects.** **Co-medication with a STRONG CYP3A INHIBITOR:** **1. Choose an alternative if possible.** **2. If an alternative is not an option: consider a dose of 84 mg eliglustat 1x daily and be alert to side effects.** **Co-medication with a MODERATE CYP3A INHIBITOR:** **1. Choose an alternative.** **2. If an alternative is not an option: consider a dose of 84mg eliglustat 1x daily and be alert to side effects.** **Co-medication with a STRONG CYP3A INDUCER: Eliglustat is not recommended. The plasma concentration may decrease so sharply that a therapeutic effect cannot be achieved.** **1. Choose an alternative if possible.**
FDA	CYP2D6 IMs: 84 mg orally 2x daily. Co-administration with strong CYP3A inhibitors is contraindicated in intermediate and poor CYP2D6 metabolizers.

More information about this drug-phenotype can also be found at **EMA**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Erdafitinib

Balversa, Erdafitinib

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Escitalopram || CYP2C19

Cipralex, Depram, Elxion, Escipra, Escitalopram, Escitalopram Oxalate, Lepax, Lexapro

START | STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Consider a clinically appropriate alternative antidepressant not predominantly metabolized by CYP2C19.
Increased metabolism of citalopram and escitalopram to less active compounds when compared to CYP2C19 rapid and normal metabolizers. Lower plasma concentrations decrease the probability of clinical benefit.

Recommendations based on specific guidelines

CPIC	**Consider a clinically appropriate alternative antidepressant not predominantly metabolized by CYP2C19. If citalopram or escitalopram are clinically appropriate, and adequate efficacy is not achieved at standard maintenance dosing, consider titrating to a higher maintenance dose. Drug-drug interactions and other patient characteristics (e.g., age, renal function, liver function) should be considered when adjusting dose or selecting an alternative therapy.**
DPWG	Avoid escitalopram. Choose drugs that are not metabolized or that are metabolized to a lesser extent by CYP2C19.
FDA	No actionable recommendation available for this drug-gene pair.

Caveat

Patients on a stable and effective dose of an SSRI most likely will not benefit from additional dose modifications based on CYP2D6 or CYP2C19 genotype results. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

28614176

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Escitalopram || CYP2D6

Cipralex, Depram, Elxion, Escipra, Escitalopram, Escitalopram Oxalate, Lepax, Lexapro

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is not a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**
FDA	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Esomeprazole

Arcolase, Axiago, Depump, Emanera, Emazole, Emp, Epivetier, Esmozin, Esoferr, Esola, Esomax, E-Some, Esomeprazole, Esomeprazole Sandoz, Esoz, Esozid, Exocid, Ezocon, Ezol, Ezomeb, Jubium, Lanxium, Nexigas, Nexium, Nexium Mups, Proxium, S - Omevell, Simprazol, Sompraz

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The CPIC Dosing Guideline for CYP2C19 and Proton Pump Inhibitor Dosing states that inconsistent findings regarding the effect of CYP2C19 genotype on the pharmacokinetics and therapeutic response to esomeprazole preclude making recommendations for these proton pump inhibitors.

Recommendations based on specific guidelines

CPIC	**No actionable recommendation available for this drug-gene pair.**
DPWG	No actionable recommendation available for this drug-gene pair.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

27107097 28674348 26730167

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Fenofibrate and Simvastatin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Fesoterodine
Fesoterodine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source
Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Flecainide

Flecainide, Tambocor

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Indications other than diagnosis of Brugada syndrome: reduce the dose to 75% of the standard dose and record an ECG and monitor the plasma concentration.
The genetic variation reduces conversion of flecainide to inactive metabolites. This may increase the risk of side effects.

Recommendations based on specific guidelines

DPWG	**Indications other than diagnosis of Brugada syndrome: reduce the dose to 75% of the standard dose and record an ECG and monitor the plasma concentration. Provocation test for diagnosis of Brugada syndrome: No action required. At a dose of 2.0 mg/kg body weight to a maximum of 150 mg, the response is better for patients with alleles that result in reduced activity. All 5 patients with these alleles and 20% of the patients with two fully active alleles exhibited a response within 30 minutes.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Flibanserin || CYP2C19

Flibanserin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Flibanserin || CYP2C9

Flibanserin

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In 8 women who were poor metabolizers of CYP2C9, Cmax and AUC0-inf of flibanserin 100 mg once daily decreased 23% and 18%, compared to exposures among 8 extensive metabolizers of CYP2C9.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Flibanserin || CYP2D6

Flibanserin

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In 12 poor metabolizers of CYP2D6, steady state Cmax and AUC of flibanserin 50 mg twice daily was decreased by 4% and increased by 18%, respectively, compared to exposures among 19 extensive metabolizers, intermediate metabolizers and ultrarapid metabolizers of CYP2D6.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Fluoxetine

Antiprestine, Apo-Fluoxetine, Deprezac, Doxetin, Elizac, Fluoxetine, Fluoxetine-Teva, Fluoxone Divule, Foransi, Kalxetin, Magrilan, Nopres, Noxetine, Pms-Fluoxetine, Prestin, Proctin, Prozac, Sactine, Zac

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Decreased metabolism of fluoxetine and increased fluoxetine:norfluoxetine ratio but similar total active enantiomer concentrations compared to normal metabolizers.

Recommendations based on specific guidelines

CPIC	**No actionable recommendation available for this drug-gene pair.**
DPWG	NO action is needed for this gene-drug interaction.

Caveat

Patients on stable and effective antidepressant medication doses without significant tolerability concerns may not benefit from dose modifications based on CYP2D6, CYP2C19, and/or CYP2B6 genotype results. Pharmacogenetic test results are one of many pieces of clinical information to be considered when optimizing antidepressant drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Flupenthixol

Fluanxol, Flupenthixol

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction. No studies have been published in which the kinetics and the effects of flupentixol were studied for this phenotype.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Flurbiprofen

Flurbiprofen, Froben, Strepfen

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Initiate therapy with recommended starting dose.
Normal metabolism.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.**

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Fluvastatin

Fluvastatin, Lescol

START STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Prescribe desired starting dose and adjust doses of fluvastatin based on disease-specific guidelines.

CYP2C9 Normal function: Normal exposure. SLCO1B1 Normal metabolizer: Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses of fluvastatin based on disease-specific guidelines.**

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Fluvastatin || CYP2C9

Fluvastatin, Lescol

START

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Prescribe desired starting dose and adjust doses of fluvastatin based on disease-specific guidelines.

Normal exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses of fluvastatin based on disease-specific guidelines.**

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Fluvastatin || SLCO1B1

Fluvastatin, Lescol

START

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses based on disease-specific guidelines.**

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation

This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Fluvoxamine || CYP2C19

Apo-Fluvoxamine, Faverin, Fluvoxamine, Luvox

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is not a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Fluvoxamine || CYP2D6

Apo-Fluvoxamine, Faverin, Fluvoxamine, Luvox

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Initiate therapy with recommended starting dose.

Reduced metabolism when compared to extensive metabolizers. Higher plasma concentrations may increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose.**
DPWG	No actionable recommendation available for this drug-gene pair.

Caveat

Patients on stable and effective antidepressant medication doses without significant tolerability concerns may not benefit from dose modifications based on CYP2D6, CYP2C19, and/or CYP2B6 genotype results. Pharmacogenetic test results are one of many pieces of clinical information to be considered when optimizing antidepressant drug therapy

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Formoterol || CYP2C19

Atimos, Duaklir Genuair, Foradil, Formoterol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Formoterol || CYP2D6

Atimos, Duaklir Genuair, Foradil, Formoterol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Some patients may be deficient in CYP2D6 or 2C19 or both. Whether a deficiency in one or both of these isozymes results in elevated systemic exposure to formoterol or systemic adverse effects has not been adequately explored.

Recommendations based on specific guidelines

FDA	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Fosphenytoin

Fosphenytoin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

CYP2C9 plays the major role in the metabolism of phenytoin.

Recommendations based on specific guidelines

HCSC	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Galantamine

Galantamine, Reminyl

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

31064198

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Gefitinib

Actagef, Gefinib, Gefitero, Gefitinib, Gefiza, Genessa, Ingefitinib, Iressa, Iretinib

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

NO action is needed.

Side effects can occur more frequently, as the gene variation increases the gefitinib plasma concentration. However, the side effects are reversible and manageable, to an extent that adjustment of the therapy in advance is not necessary.

Recommendations based on specific guidelines

DPWG	NO action is needed.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Glibenclamide || CYP2C9

Apo-Glyburide, Benil, Clamide, Daonil, Dibelet, Fimediab, Glibenclamide, Glimide, Glyboral, Harmida, Hisacha, Latibet, Prodiabet, Renabetic, Sp-Diatab, T.O.Nil, Trodeb

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Gliclazide || CYP2C9

Apo-Gliclazide, Diamicron, Dianorm, Diapro, Diverin, Fonylin, Gliavis, Glicab, Gliclada, Gliclazide, Glidex, Glikamel, Glimicron, Glizide, Glucodex, Glucored, Glukolos, Glyade, Glyclazide, Gored, Linodiab, Medoclazide, Melicron, Mexan, Pedab, Sp-Glimed, Sun-Glizide, Xepabet

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Glimepiride || CYP2C9

Amaglu, Amaryl, Amidiab, Anpiride, Dialosa, Diapride, Diaversa, Friladar, Glamarol, Gliaride, Glimefion, Glimepiride, Glimepix, Glimetic, Glucokaf, Glucoryl, Gluvas, Metrix, Norizec, Relide, Simryl, Velacom, Versibet

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Haloperidol

Apo-Haloperidol, Dores, Govotil, Haloperidol, Haloxen, Lodomer, Manace, Motivan, Myung In Haloperidol, Seradol, Upsikis

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The genetic variation results in a higher plasma concentration, but the effect is small and no clinically significant effects were found.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Hydrocodone

Hydrocodone

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Use hydrocodone label recommended age- or weight-specific dosing.
Minimal evidence for pharmacokinetic or clinical effect.

Recommendations based on specific guidelines

CPIC	**Use hydrocodone label recommended age- or weight-specific dosing. If no response and opioid use is warranted, consider non-codeine or non-tramadol opioid.**

Caveat

Like all diagnostic tests, CYP2D6 genotype is one of multiple pieces of information that clinicians should consider in guiding their therapeutic choice for each patient. Furthermore, there are several other factors that cause potential uncertainty in the genotyping results and phenotype predictions.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Ibrutinib

Ibrutinib, Imbruvica

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Ibuprofen

Xepafen, Zentarin, Zofen, Advil, Anafen, Apo-Ibuprofen, Arbupon, Arfen, Arthrifen, Axofen, Bifen, Bodrexin Ibp, Brufen, Bufect, Bunofa, Caldolor, Coldy's Jr, Dofen Forte, Dolofen F, Dratadin, Etafen, Etafen Forte, Farsifen, Febryn, Fenatic, Fenida, Fenpro, Fenris, Hufagripp, Ibrosic, Ibufen, Ibuleve, Ibuprofen, Ifen, Insic, Intrafen, Iprox, Irgafen, Junifen, Lexaprofen, Liflamal, Medcofen, Mediprofen, Mofen, Moris, Neo Linucid, Novaxifen, Nurofen, Oraprofen, Ostarin, Panafen, Peinlos, Perofen, Profen, Prointi, Proris, Proris Forte, Prosinal, Prosinal Forte, Provinas, Repass, Rhelafen, Ribunal, Salfenal, Simris, Spedifen, Termofen, Tiafen, Trobuges

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Initiate therapy with recommended starting dose.
Normal metabolism.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.**

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Iloperidone

Iloperidone

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Imipramine

Apo-Imipramine, Imipramine

START **MODERATE RECOMMENDATION**

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider alternative drug not metabolized by CYP2C19.

CYP2D6 IM: Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds. Higher plasma concentrations of active drug may increase probability of side effects. CYP2C19 UM: Increased metabolism of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Consider alternative drug not metabolized by CYP2C19. If tricyclic antidepressants (TCAs) is warranted, utilize therapeutic drug monitoring to guide dose adjustment.**

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Imipramine || CYP2C19

Apo-Imipramine, Imipramine

START **MODERATE RECOMMENDATION**

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19.

Increased metabolism of tertiary amines compared to normal metabolizers. Greater conversion of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19. Tricyclic antidepressants (TCAs) without major CYP2C19 metabolism include the secondary amines nortriptyline and desipramine. If a tertiary amine is warranted, utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	No action is required for this gene-drug interaction.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Imipramine || CYP2D6

Apo-Imipramine, Imipramine

START MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider 25% reduction of recommended starting dose.

Reduced metabolism of tricyclic antidepressants (TCAs) to less active compounds when compared to normal metabolizers. Higher plasma concentrations of active drug will increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Consider 25% reduction of recommended starting dose. Patients may receive an initial low dose of a tricyclic, which is then increase over several days to the recommended steady-state dose. Utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	Use 70% of the standard dose and monitor the side effect or the plasma concentrations of imipramine and desipramine. Monitoring is done in order to set the maintenance dose. The therapeutic range is 150-300 ng/mL for the sum of the imipramine and desipramine plasma concentrations. Values exceeding 500 ng/mL are considered toxic.
FDA	No actionable recommendation available for this drug-gene pair.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Indomethacin

Indomethacin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No recommendation is available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	**No recommendation is available for this drug-gene pair.**

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Lacosamide

Lacosamide, Vimpat

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Lansoprazole

Caprazol, Digest, Dobrizol, Erphalanz, Inazol, Inhipraz, Lagas, Lancid, Lanpracid, Lansoprazole, Lanzogra, Lapraz, Lasgan, Lasoprol, Laz, Lexid, Loprezol, Nufaprazol, Prazotec, Prevacid, Prosogan, Prosogan Fd, Zolesco

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.
Decreased plasma concentrations of PPIs compared with CYP2C19 NMs; increase risk of therapeutic failure.

Recommendations based on specific guidelines

CPIC	**Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.**
DPWG	For Helicobacter pylori ERADICATION THERAPY: 1. Use a 4-fold higher dose. 2. Advise the patient to contact their doctor if symptoms of dyspepsia persist. OTHER INDICATIONS: 1. Be alert to reduced effectiveness. 2. If necessary, use a 4-fold higher dose. 3. Advise the patient to report persisting symptoms of dyspepsia.

Caveat

There are important limitations to CYP2C19 genetic tests. Targeted genotyping tests focus on known star (*) alleles and are not designed to detect novel variants. Rare allelic CYP2C19 variants may not be included in the genotype test, and patients with these variants may be assigned an NM phenotype (CYP2C19*1/*1) by default. An assigned *1 allele could harbor an undetected CYP2C19 variant that alters metabolism and drug exposure. Rare alleles with gene deletions at the CYP2C19 locus (*36 and *37) have been reported; however, most clinical laboratories do not test for CYP2C19 copy number variants or deletions. Clinical providers must understand the limitations of targeted genotyping and which CYP2C19 variant alleles were tested when interpreting results. CYP2C19 genotype is just one factor clinicians should consider when prescribing PPIs.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Increase Starting Dose
Patient has altered metabolism rate or reduced activation rate for drug indicated. Increased starting dose has shown to help patient into target therapeutic range faster and/or reduce the risk of drug clinical inefficacy.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Lesinurad

Lesinurad

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Letermovir

Letermovir

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

No actionable recommendation available for this drug-gene pair.

The effect of genetic variants in the OATP1B1 gene SLCO1B1 (rs4149056, rs2306283, rs4149032) and UGT1A1 (rs4148323 and promoter TA repeat variants) on the pharmacokinetics of Letermovir was evaluated in 299 study participants. There were no clinically relevant effects of these variants on Letermovir exposure.

Recommendations based on specific guidelines

SWISSMEDIC **No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Lofexidine
Lofexidine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source
Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Lornoxicam

Lornoxicam

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Initiate therapy with recommended starting dose.

Normal metabolism.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.**

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Losartan || CYP2C9

Acetensa, A-Losartan, Angioten, Angizaar, Cozaar, Hyperten, Insaar, Lifezar, Lortan, Losagen, Losargard, Losartan, Losartan Bluepharma, Losartan Hexal, Losartan Potassium, Losartas, Lozarsin, Myotan, Rosart, Santesar, Sartocad, Trosan

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Approximately 14% of a perorally administered dose of losartan is converted to an active metabolite. In vitro studies show that cytochrome P450 2C9 and 3A4 are involved in the conversion of losartan into its metabolites.

Recommendations based on specific guidelines

SWISSMEDIC **No actionable recommendation available for this drug-gene pair.**

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

33509019

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Lovastatin

Apo-Lovastatin, Cholvastin, Ellanco, Elstatin, Justin, Lochol, Loctin, Lofacol, Lotyn, Lovastatin, Lovatin, Medostatin, Minipid, Rovacor

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.
Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses based on disease-specific guidelines.**

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Lumiracoxib

Lumiracoxib

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No recommendation is available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	No recommendation is available for this drug-gene pair.

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Meclizine

Meclizine

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Monitor for adverse reactions and clinical effect.

May affect systemic concentrations.

Recommendations based on specific guidelines

FDA	Monitor for adverse reactions and clinical effect.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Increase Monitoring

Although patient has a varied genetic profile, the intended medication can still be prescribed as recommended, but with more frequent monitoring for quick action to be taken in the event of unwanted reactions.

Meloxicam

Arimed, Arrox, Artrilox, Atrocox, Cameloc, Coxilab, Denilox, Flamoxi, Flasicox, Fri-Art, Futamel, Genxicam, Hexcam, Hufaxicam, Koniflam, Liloxicam, Loxicox, Loxil, Loximei, Mecox, Meflam, Melet, Melfion, Melicam, Melocid, Melogra, Melovix, Melox, Meloxicam, Meloxin, Mevilox, Mexpharm, Mobic, Mobiflex, Movi-Cox, Movix, Moxam, Moxic, Nucoxi, Nufaxicam, Nulox, Ostelox, Oxcam, Relox, Remelox, Velcox, X-Cam

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Initiate therapy with recommended starting dose.
Normal metabolism.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.**
HCSC	No actionable recommendation available for this drug-gene pair.

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

26774055 29024493

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Methadone || CYP2D6

Methadone

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

No effect or insufficient evidence for methadone adverse events, opioid dose requirements, or analgesia.

Recommendations based on specific guidelines

CPIC	No actionable recommendation available for this drug-gene pair.

Caveat

Like all diagnostic tests, CYP2D6 genotype is one of multiple pieces of information that clinicians should consider in guiding their therapeutic choice for each patient. Furthermore, there are several other factors that cause potential uncertainty in the genotyping results and phenotype predictions.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Methotrexate

Abitrexate, Dbl Methotrexate, Ebetrexat, Emthexate, Ferxate, Kemotrexate, Methotrexate, Methotrexate Ebewe"", Metoject, Rheu-Trex, Sanotrexate, Trexan

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

No actionable recommendation available for this drug-gene pair.

Several polymorphisms of the SLCO1B1 gene, notably rs4149081 and rs11045879 have been strongly associated for the first time with methotrexate clearance and gastrointestinal toxicity.

Recommendations based on specific guidelines

PRO	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Methylphenidate

Concerta, Medikinet Mr, Methylphenidate, Prohiper, Ritalin, Rubifen

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Metoclopramide || CYP2D6

Apo-Metoclop, Damaben, Emeliv, Emeran, Ethiferan, Gavistal, Hufaclop, Maril, Metoclopramide, Nausile, Navoren, Nilatika, Norvom, Omevomid, Primpen, Primperan, Pulin, Sotatic, Syntomide, Tivomit, Tomit, Vertivom, Vopram, Vosea, Xylastin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Metoprolol

Apo-Metoprolol, Betaloc Zok, Denex, Fapresor, Lopresor, Loprolol, Metoprolol

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Initiate standard dosing.

Decreased metabolism of metoprolol leading to increased drug concentrations; however, this does not have any clinically significant changes in heart rate, blood pressure, or clinical outcomes.

Recommendations based on specific guidelines

CPIC	**Initiate standard dosing.**
DPWG	GRADUAL REDUCTION in HEART RATE or SYMPTOMATIC BRADYCARDIA: prescribe no more than 50% of the standard dose. The use of smaller steps in dose titration can also be done together or as another option. For OTHER CASES: no action required.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Mirabegron

Betmiga, Mirabegron

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Mirtazapine || CYP2C19

Apo-Mirtazapine, Mirtazapine, Mirtazapine Sandoz, Mirzap, Remeron Soltab

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Mirtazapine || CYP2D6

Apo-Mirtazapine, Mirtazapine, Mirtazapine Sandoz, Mirzap, Remeron Soltab

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The higher plasma concentration of mirtazapine does not result in an increase in the side effects.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

31158907 26595747 25475885

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Moclobemide

Moclobemide, Moclobemide Hexal

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Although the moclobemide plasma concentration may decrease as a result of increased CYP2C19 activity, this does not lead to increased effectiveness, as far as is known.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Modafinil

Modafinil

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Nabumetone

Goflex

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No recommendation is available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	No recommendation is available for this drug-gene pair.

Caveat

Rare CYP2C9 variants may not be detected in genotype tests, classifying patients with these variants as having a normal metabolizer (NM) phenotype based on CYP2C9*1/1 results. The CYP2C9*1 allele may harbor reduced or non-functional variants, so reports should specify the variants or SNPs analyzed. CYP2C9 genotype is one of many factors to consider when prescribing NSAIDs. Age, sex, ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI risk. Use with caution in elderly patients, as they have reduced CYP2C9 metabolism and increased renal/GI risks. Combining NSAIDs with antiplatelets or anticoagulants raises bleeding risk or can interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Naproxen

Alif, Xenifar

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The pharmacokinetics of this drug is not significantly impacted by CYP2C9 genetic variants in vivo and/or there is insufficient evidence to provide a recommendation to guide clinical practice at this time.

Recommendations based on specific guidelines

CPIC	No actionable recommendation available for this drug-gene pair.

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Nateglinide

Nateglinide, Starlix Tablet

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Data available from both in vitro and in vivo experiments indicate that nateglinide is predominantly metabolized by CYP2C9 with involvement of CYP3A4 to a smaller extent.

Recommendations based on specific guidelines

EMA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Nebivolol

Linoven, Nebilet, Nebivolol, Nevodio

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Nebivolol is metabolized via alicyclic and aromatic hydroxylation, N-dealkylation and glucuronidation, partly to active metabolites. Although the aromatic hydroxylation is part of the CYP2D6-dependent genetic oxidative polymorphism, the active metabolites achieve a similar effect in the rapidly and slowly metabolizing patients.

Recommendations based on specific guidelines

SWISSMEDIC **No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Nefazodone

Nefazodone

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Nelfinavir

Nelfinavir

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Nelfinavir is metabolized by CYP3A and CYP2C19. Coadministration of nelfinavir and drugs that induce CYP3A or CYP2C19, such as rifampin, may decrease nelfinavir plasma concentrations and reduce its therapeutic effect. Coadministration of nelfinavir and drugs that inhibit CYP3A or CYP2C19 may increase nelfinavir plasma concentrations.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Nortriptyline

Apo-Nortriptyline, Nortriptyline

START MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider 25% reduction of recommended starting dose.

Reduced metabolism of TCAs to less active compounds when compared to normal metabolizers. Higher plasma concentrations of active drug will increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Consider 25% reduction of recommended starting dose. Patients may receive an initial low dose of a tricyclic, which is then increase over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose. Utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
DPWG	Use 60% of the standard dose. Monitor effect and side effects or plasma concentration to set maintenance dose. The therapeutic range of nortriptyline is 50-150 ng/mL. Values exceeding 250 ng/mL are considered toxic.
FDA	No actionable recommendation available for this drug-gene pair.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Olanzapine || CYP2D6

Apo-Olanzapine, Jubrexa, Olancor Fc, Olankline, Olanzapine, Olanzapine Mevon, Olanzapine Stada, Olanzavitae, Olaz, Olzan, Olzanid, Onzapin, Prolanza, Remital, Sopavel, Tolanz, Zypin, Zyprexa, Zyprexa Im, Zyprexa Zydis

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

26856397

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Oliceridine
Oliceridine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Omeprazole

Etagastrin, Gastrazol, Gastrofer, Inhipump, Lanacer, Lokev, Losec, Losec Mups, Meisec, Norsec, Ocid, Ofizol, Olit, Omberzol, Omed, Omenole, Omepradex, Omeprazole, Omeprazole Kabi, Omeprazole Sandoz, Omesec, Omevell, Omeyus, Omezol Lyo-, Omezole, Omicap, Omz, Penrazol, Probitor Gastro-Resistant, Proceptin, Promesec, Protop, Pumpitor, Ramezol, Redusec, Rindopump, Rocer, Romesec, Solcer, Ulpraz Zolacap, Zenpro, Zimor, Zollocid

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.
Decreased plasma concentrations of PPIs compared with CYP2C19 NMs; increase risk of therapeutic failure.

Recommendations based on specific guidelines

CPIC	**Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.**
DPWG	For Helicobacter pylori ERADICATION THERAPY: 1. use a 3-fold higher dose. 2. advise the patient to contact their doctor if symptoms of dyspepsia persist. OTHER INDICATIONS: 1. be alert to reduced effectiveness. 2. if necessary, use a 3-fold higher dose. 3. advise the patient to report persisting symptoms of dyspepsia.

Caveat

There are important limitations to CYP2C19 genetic tests. Targeted genotyping tests focus on known star (*) alleles and are not designed to detect novel variants. Rare allelic CYP2C19 variants may not be included in the genotype test, and patients with these variants may be assigned an NM phenotype (CYP2C19*1/*1) by default. An assigned *1 allele could harbor an undetected CYP2C19 variant that alters metabolism and drug exposure. Rare alleles with gene deletions at the CYP2C19 locus (*36 and *37) have been reported; however, most clinical laboratories do not test for CYP2C19 copy number variants or deletions. Clinical providers must understand the limitations of targeted genotyping and which CYP2C19 variant alleles were tested when interpreting results. CYP2C19 genotype is just one factor clinicians should consider when prescribing PPIs.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

25498969 24996380 27107097

Increase Starting Dose
Patient has altered metabolism rate or reduced activation rate for drug indicated. Increased starting dose has shown to help patient into target therapeutic range faster and/or reduce the risk of drug clinical inefficacy.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Ondansetron

Cendatron, Ceteron, Dansefion, Dantroxal, Entron, Frazon, Fudanton, Gencetron, Glotron, Imorfoz, Insetron, Invomit, Kliran, Lametic, Maxtron, Mefoz, Narfoz, Natzo, Nausimex, Odanostin, Odnatron, ODR, Ondacap, Ondamet, Ondane, Ondansetron, Ondansetron Hexal, Ondansetron Kabi, Ondansetron Sandoz, Ondansetron-Aft, Ondarin, Ondaster, Ondavar, Ondavell, Ondero, Ondesco, Onetic, Onron, Rivomet, Setronax, Sydnatron, Trisetron, Tronadex, Trovensis, Vastercon, Vomceran, Vometraz, Vometron, Vomigo, Vondatum, Zetral, Zetron, Zofran

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Initiate therapy with recommended starting dose. Insufficient evidence showing clinical impact based on CYP2D6 genotype.
Very limited data available for CYP2D6 intermediate metabolizers.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. Insufficient evidence showing clinical impact based on CYP2D6 genotype.**
FDA	No actionable recommendation available for this drug-gene pair.

More information about this drug-phenotype can also be found at **SWISSMEDIC**.

Caveat

Rare CYP2D6 variants may not be included in the genotype test used and patients with rare variants may be assigned a “wildtype” (CYP2D6*1) genotype by default. Thus, an assigned “wildtype” allele could potentially harbor a no or decreased function variant. Furthermore, it is important that the genetic testing platform include testing for gene copy number to identify CYP2D6 UMs. Caution should be used regarding molecular diagnostics of CYP2D6 gene copy-number variation because commercially available genotyping results may differ between diagnostic laboratories depending on assay design. Like all diagnostic tests, CYP2D6 genotype is one of multiple pieces of information that clinicians should consider when making their therapeutic choice.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Ospemifene || CYP2C9

Ospemifene, Sensio

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Ospemifene is metabolized primarily by CYP3A4 and CYP2C9. CYP2C19 and other pathways contribute to the metabolism of ospemifene. In order of decreasing potency, ospemifene was suggested to be a weak inhibitor for CYP2B6, CYP2C9, CYP2C19, CYP2C8, CYP2D6 and CYP3A4 in in vitro studies.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

More information about this drug-phenotype can also be found at **HCSC**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Oxycodone

Oxycodone, Oxycodone Wockhardt, Oxycontin, Oxycontin Neo, Oxyneo, Oxynorm, Targin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No recommendation for oxycodone therapy because of weak evidence regarding adverse events or analgesia.

Decreased metabolism of oxycodone to active metabolite oxymorphone, but this does not appear to translate into decreased analgesia or side effects.

Recommendations based on specific guidelines

CPIC	**No recommendation for oxycodone therapy because of weak evidence regarding adverse events or analgesia.**
DPWG	No actionable recommendation available for this drug-gene pair.

Caveat

Like all diagnostic tests, CYP2D6 genotype is one of multiple pieces of information that clinicians should consider in guiding their therapeutic choice for each patient. Furthermore, there are several other factors that cause potential uncertainty in the genotyping results and phenotype predictions.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Paliperidone

Invega, Invega Sustenna, Invega Trinza, Paliperidone

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No adjustment in the paliperidone dose on the basis of predicted phenotype is warranted.

Recommendations based on specific guidelines

HCSC	**No adjustment in the paliperidone dose on the basis of predicted phenotype is warranted. This is based on a population pharmacokinetic analysis to evaluate the influence of predicted CYP2D6 phenotype on exposure.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Palonosetron

Aloxi, Palofer, Palonosetron, Paloset, Paloxi, Palset, Prosmol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Dose adjustment is not necessary.

The genetic polymorphism CYP2D6 does not significantly influence the overall body clearance of palonosetron compared to healthy individuals.

Recommendations based on specific guidelines

SWISSMEDIC **Dose adjustment is not necessary.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Pantoprazole

Alpanzole, Caprol, Ciprazol, Controloc, Duonazole, Erprazol, Panloc, Panso, Pantera, Pantin, Pantocapri, Pantomet, Pantomex, Pantoprazole, Pantoprazole Normon, Pantoprazole Sandoz, Pantopump, Panvell, Panzole, Panzolec, Peptazol, Pepzol, Pranza, Prazopump, Pumpisel, Topazol, Vomizole

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.

Decreased plasma concentrations of PPIs compared with CYP2C19 NMs; increase risk of therapeutic failure.

Recommendations based on specific guidelines

CPIC	**Increase starting daily dose by 100%. Daily dose may be given in divided doses. Monitor for efficacy.**
DPWG	For Helicobacter pylori ERADICATION THERAPY: 1. Use a 5-fold higher dose. 2. Advise the patient to contact their doctor if symptoms of dyspepsia persist. OTHER INDICATIONS: 1. Be alert to reduced effectiveness. 2. If necessary, use a 5-fold higher dose. 3. advise the patient to report persisting symptoms of dyspepsia.

Caveat

There are important limitations to CYP2C19 genetic tests. Targeted genotyping tests focus on known star (*) alleles and are not designed to detect novel variants. Rare allelic CYP2C19 variants may not be included in the genotype test, and patients with these variants may be assigned an NM phenotype (CYP2C19*1/*1) by default. An assigned *1 allele could harbor an undetected CYP2C19 variant that alters metabolism and drug exposure. Rare alleles with gene deletions at the CYP2C19 locus (*36 and *37) have been reported; however, most clinical laboratories do not test for CYP2C19 copy number variants or deletions. Clinical providers must understand the limitations of targeted genotyping and which CYP2C19 variant alleles were tested when interpreting results. CYP2C19 genotype is just one factor clinicians should consider when prescribing PPIs.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Increase Starting Dose
Patient has altered metabolism rate or reduced activation rate for drug indicated. Increased starting dose has shown to help patient into target therapeutic range faster and/or reduce the risk of drug clinical inefficacy.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Paracetamol, Caffeine, and Dihydrocodeine

Paracetamol, Caffeine, and Dihydrocodeine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Paracetamol, Combinations Excl. Psycholeptics

Paracetamol, Combinations Excl. Psycholeptics

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Paroxetine

Apo-Paroxetine, Paroxetine, Seroxat

START **MODERATE RECOMMENDATION**

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider a lower starting dose and slower titration schedule as compared to normal metabolizers.
Reduced metabolism of paroxetine to less active compounds when compared to CYP2D6 normal metabolizers when starting treatment or at lower doses.

Recommendations based on specific guidelines

CPIC	**Consider a lower starting dose and slower titration schedule as compared to normal metabolizers. Drug-drug interactions and other patient characteristics (e.g., age, renal function, liver function) should be considered when adjusting dose or selecting an alternative therapy.**
DPWG	No action is needed for this gene-drug interaction.
FDA	No actionable recommendation available for this drug-gene pair.

Caveat

Patients on a stable and effective dose of an SSRI most likely will not benefit from additional dose modifications based on CYP2D6 or CYP2C19 genotype results. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

24868171

Decrease Starting Dose
Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Perphenazine

Apo-Perphenazine, Perphenazine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Phenprocoumon || CYP2C9

Phenprocoumon

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Phenytoin || CYP2C19

Curelepz, Dbl Phenytoin, Decatona, Dextoin, Dilantin, Ikaphen, Kutoin, Phenitin, Phenytoin, Sanbetoin

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Phenytoin is metabolized by the cytochrome P450 enzymes CYP2C9 and CYP2C19.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Phenytoin || CYP2C9

Curelepz, Dbl Phenytoin, Decatona, Dextoin, Dilantin, Ikaphen, Kutoin, Phenitin, Phenytoin, Sanbetoin

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Metabolism: Phenytoin is more than 95% biotransformed in the liver (mostly via CYP2C9 and to a small extent via CYP2C19).

Recommendations based on specific guidelines

SWISSMEDIC **No actionable recommendation available for this drug-gene pair.**

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

25096692 30270535 31646624

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Pimozide

Pimozide

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Use no more than 80% of the standard maximum dose: 16 mg/day (12 years and older).

The risk of QT-prolongation – and thereby also the risk of torsade de points – is theoretically increase, because the genetic variation results in an increase in the plasma concentration of pimozide. The risk of an excessively high plasma concentration can be negated by following the dose recommendations provided.

Recommendations based on specific guidelines

DPWG	Use no more than 80% of the standard maximum dose: 16 mg/day (12 years and older).

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose

Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Piroxicam

Apo-Piroxicam, Aritmatic, Axcel, Benoxicam, Campain, Capxidin, Denicam, Faxiden, Feldco, Feldene, Flaxicam, Fleroxi, Genroxi, Grazeo, Infeld, Inpirox, Kifadene, Lanareuma, Lexicam, Licofel, Maxicam, Mesaden, Miradene, Novaxicam, Omeretik, Ovtelis, Pirocam, Pirofel, Piroxicam, Rheficam, Rhumagel, Robilex, Rosic, Rosiden, Roxidene, Roxifen, Samrox, Scandene, Selmatic, Triadene, Tropidene, Wiros, Xicalom, Xicam, Yasiden

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Initiate therapy with recommended starting dose.
Normal metabolism.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.**

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Pitavastatin

Livalo, Pitavastatin, Vitor

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses based on disease-specific guidelines.**

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Pitolisant
Pitolisant

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Prasugrel || CYP2C19

Effient, Prasugrel

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**
FDA	No actionable recommendation available for this drug-gene pair.

More information about this drug-phenotype can also be found at **EMA** and **SWISSMEDIC**.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

32664772 30092595

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Prasugrel || CYP2C9

Effient, Prasugrel

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In healthy subjects, patients with stable atherosclerosis, and patients with ACS receiving prasugrel, there was no relevant effect of genetic variation in CYP2B6, CYP2C9, CYP2C19, or CYP3A5 on the pharmacokinetics of prasugrel's active metabolite or its inhibition of platelet aggregation.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

More information about this drug-phenotype can also be found at **EMA** and **SWISSMEDIC**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Pravastatin

Apo-Pravastatin, Cholespar, Gravastin, Koleskol, Meprastin, Mevachol, Novales, Novosta, Prafaven, Pravastatin, Pravinat

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses based on disease-specific guidelines.**

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Propafenone

Propafenone, Rytmonorm

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Either guide the dose by therapeutic drug monitoring, perform an ECG and be alert to side effects or choose an alternative.

Genetic variation increases the sum of the plasma concentrations of propafenone and the active metabolite 5-hydroxypropafenone. This may increase the risk of side effects.

Recommendations based on specific guidelines

DPWG	**Either guide the dose by therapeutic drug monitoring, perform an ECG and be alert to side effects or choose an alternative. Antiarrhythmic drugs that are hardly if at all metabolized by CYP2D6 include, for example, sotalol, disopyramide, quinidine and amiodarone. It is not possible to offer adequately substantiated recommendations for dose adjustment based on the literature.**
SWISSMEDIC	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription

There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Propranolol

Apo-Propanolol, Emforal, Farmadral, Inpanol, Liblok, Nalol, Propa, Propanol, Propranolol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Protriptyline

Protriptyline

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Quetiapine || CYP2D6

Alvoquel, Apo-Quetiapine, Ketipinor, Q-Pin, Quetiapine, Quetiapine Sandoz, Quetvell, Seroquel, Soroquin

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Quinidine

Quinidine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Quinine || CYP2D6

Kina, Quinine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Rabeprazole

Acilesol, Barole, Bepraz, Pariet, Rabeprazole, Rabeprazole Sandoz

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Inconsistent findings regarding the effect of CYP2C19 genotype on the pharmacokinetics and therapeutic response to rabeprazole preclude making recommendations for these second-generation PPIs.

Recommendations based on specific guidelines

CPIC	**No actionable recommendation available for this drug-gene pair.**
DPWG	No actionable recommendation available for this drug-gene pair.
PMDA	No actionable recommendation available for this drug-gene pair.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

28674348

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Ranolazine

Ranexa, Ranolazine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Rimegepant

Rimegepant

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

CYP2C9 activity is reduced in individuals with genetic variants such as the CYP2C9*2 and CYP2C9*3 alleles.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

More information about this drug-phenotype can also be found at **EMA**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Risperidone

Apo-Risperidone, Eperon, Neripros, Noprenia, Nuperdal, Persidal, Respirex, Ridal, Ridkline, Rispator, Rispefar, Risperdal, Risperdal Consta, Risperidex, Risperidone, Risperidone Mevon, Risperidone-Chanelle, Rizodal

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

There is little evidence to support an increase in side effects caused by the genetic variation. The genetic variation may lead to a decrease in the required maintenance dose. However, as the effect on the dose is smaller than that of the normal biological variation, action is not useful.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

24828442

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Ritonavir

Norvir, Ritonavir

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Rosuvastatin || SLCO1B1

Alvostat, Apo-Rosuvastatin, Celmantin, Cholostar, Crestor, Eurovastin, Giventor, Nistrol, Pyfaros, Recansa, Rosatin, Rosfion, Rosucard, Rosufer, Rosupid, Rosuvastatin, Rosuvastatin Sandoz, Rosuvastatin Winthrop, Roswin, Rovas, Rovast, Rovastar, Rovaster, Rovator, Rozact, Ruvastin, Simrovas, Suvesco, Tintaros, Vastrol, Vivacor

START STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.
Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses based on disease-specific guidelines.**

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

33150478 25630984

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Rucaparib

Rucaparib

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In vitro, rucaparib was metabolized primarily by CYP2D6 and to a lesser extent by CYP1A2 and CYP3A4. Based on population pharmacokinetic analyses, steady-state concentrations following rucaparib 600 mg twice daily did not differ significantly across CYP2D6 or CYP1A2 genotype subgroups.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

More information about this drug-phenotype can also be found at **EMA**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Sertindole

Sertindole

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Sertraline || CYP2C19

Apo-Sertraline, Deptral, Fatral, Fridep, Iglodep, Inosert, Nudep, Seratine, Serlof, Sernade, Sertraline, Setrof, Zerlin, Zoloft

START STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Initiate therapy with recommended starting dose.
Small increase in metabolism of sertraline to less active compounds when compared to CYP2C19 normal metabolizers.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. CYP2B6 metabolizer status, drug-drug interactions, and other patient characteristics (e.g., age, renal function, liver function) should also be considered.**
DPWG	No actionable recommendation available for this drug-gene pair.

Caveat

Patients on a stable and effective dose of an SSRI most likely will not benefit from additional dose modifications based on CYP2D6 or CYP2C19 genotype results. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Sertraline || CYP2D6

Apo-Sertraline, Deptral, Fatral, Fridep, Iglodep, Inosert, Nudep, Seratine, Serlof, Sernade, Sertraline, Setrof, Zerlin, Zoloft

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is not a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Simvastatin

Apo-Simvastatin, Cholestat, Colesbat, Cortavas, Covastin, Dha-Simvastatin, Esvat, Ifistatin, Kardak, Kolefion, Lesvatin, Lextatin, Lipivast, Mastatin, Mersivas, Mevastin, Norpid, Preschol, Priacin, Rechol, Rendapid, Rocoz, Selvas, Selvat, Selvim, Simbado, Simcard, Simchol, Simlo, Simtin, Simvacor, Simvas, Simvaschol, Simvasgen, Simvastal, Simvastatin, Simvasto, Simvor, Simzal, Sinova, Statcol, Statkoles, Svt, Tianvas, Valansim, Valemia, Vascor, Vasilip, Zencor, Zochol, Zocor

START

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
SLCO1B1	rs4149056(TT)	SLCO1B1 Normal Function

Recommendation

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.
Typical myopathy risk and statin exposure.

Recommendations based on specific guidelines

CPIC	**Prescribe desired starting dose and adjust doses based on disease-specific guidelines.**
PRO	Continue treatment taking into account the clinical risk. Discontinue if muscle toxicity occurs.
SWISSMEDIC	The absence of this gene in genotyping does not exclude the possibility of myopathy. Including drugs combination Ezetimibe and Simvastatin.

Caveat

Genetic variation is just one factor when prescribing statins. Rare variants may not be included in the genotype test. Patients with rare variants that reduce SLCO1B1 function may be incorrectly assigned a normal phenotype. Many patients' statin therapy is never restarted after Statin-related musculoskeletal symptoms. As a result, LDL cholesterol values are higher as is their risk for cardiovascular disease. The evidenced-based recommendations for genotype-guided statin therapy are focused on reducing the risk of SAMS.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

33150478 25932441 30336686

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

START **Initiation**
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Siponimod

Mayzent, Siponimod

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Sotalol

Apo-Sotalol, Pms-Sotalol, Sotahexal, Sotalol

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	No actionable recommendation available for this drug-gene pair.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Tamoxifen

Novaldex, Tamofen, Tamoxifen

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer (non-*10)

Recommendation

Consider aromatase inhibitor (AI) for postmenopausal or AI with ovarian function suppression for premenopausal women.

Lower endoxifen concentrations; higher risk of breast cancer recurrence, event-free, and recurrence-free survival.

Recommendations based on specific guidelines

CPIC	**Consider aromatase inhibitor (AI) for postmenopausal or AI with ovarian function suppression for premenopausal women. If AI use is contraindicated, consideration should be given to use a higher but FDA approved tamoxifen dose (40 mg/day). Avoid CYP2D6 strong to weak inhibitors.**
DPWG	Select an alternative or measure the endoxifen concentration and increase the dose if necessary by a factor of 1.5-2. Aromatase inhibitors are a possible alternative for post-menopausal women. If TAMOXIFEN is selected: avoid co-medication with CYP2D6 inhibitors such as paroxetine and fluoxetine.
PRO	No actionable recommendation available for this drug-gene pair.
FDA	The impact of CYP2D6 intermediate or poor metabolism on efficacy is not well established.

More information about this drug-phenotype can also be found at **CPNDS** and **HCSC**.

Caveat

For the treatment of ER1 breast cancer, there are well-accepted tumor somatic factors that drive endocrine response, including the tumor expression of ER, PR, and HER2 expression, and other multigene assays that are associated with endocrine sensitivity. Although there are very few data, the implication of reduced CYP2D6 metabolism in patients with low-risk breast cancer (e.g., early-stage breast cancer where the risk of distant recurrence is low) may be substantially different than in patients with later stage disease with a much higher risk of distant recurrence.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

25091503 30357449 31821071

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Tamsulosin

Farlosin Sr, Harnal D, Harnal Ocas, Prostam, Tamsin, Tamsulosin, Tasulose

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Tenoxicam

Analcam, Notritis, Pilopil, Tenoxicam, Thenil, Tilarco, Tilflam

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

Initiate therapy with recommended starting dose.

Normal metabolism.

Recommendations based on specific guidelines

CPIC	**Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.**

Caveat

Age, sex, race and ethnicity, liver dysfunction, comorbidities, concomitant medications, genetic linkage with CYP2C8, and other factors (genetic and environmental) influence adverse event risk. NSAIDs should be avoided in patients with renal dysfunction, heart failure, or high cardiovascular or GI adverse events. Usage in elderly patients must be with caution, as they have reduced CYP2C9 metabolism and increased renal risk and GI adverse events. Use extreme caution when combining NSAIDs with antiplatelets or anticoagulants as it can raise bleeding risk or interfere with platelet inhibition in the case of aspirin. NSAIDs also reduce antihypertensive efficacy, necessitating cautious use in hypertensive patients.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Terbinafine

Dermafin, Farmasil, Haterbin, Interbi, Lamisil, Lisim, Meccaderma, Solveasy Tinea, Terbinafine, Termisil

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Tetrabenazine

Tetrabenazine

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Exercise caution, closely monitor, and gradually increase the dose in these patients only if tolerated.

Patients receiving CYP2D6 inhibitors, CYP2D6 poor metabolizers deficient in CYP2D6 activity and CYP2D6 intermediate metabolizers with reduced CYP2D6 activity are at increased risk of developing side effects due to higher levels of the active metabolites of this product.

Recommendations based on specific guidelines

PMDA	**Exercise caution, closely monitor, and gradually increase the dose in these patients only if tolerated.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Increase Monitoring

Although patient has a varied genetic profile, the intended medication can still be prescribed as recommended, but with more frequent monitoring for quick action to be taken in the event of unwanted reactions.

Thioridazine

Thioridazine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Ticagrelor

Clotaire, Ticagrelor, Briclot, Brilinta

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

This is NOT a gene-drug interaction.

Recommendations based on specific guidelines

DPWG	**No actionable recommendation available for this drug-gene pair.**
EMA	No actionable recommendation available for this drug-gene pair.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

32664772 34708996 35727586

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Timolol

Duotrav, Isotic Adretor, Nyolol, Ophtamol, Opthil, Optimol, Tiamol, Timabak, Timo-Comod, Timol, Timolol, Timolol-Pos, Tim-Ophtal, Timoptol, Timoptol-Xe, Ximex Opticom

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Potentiated systemic beta-blockade (e.g., decreased heart rate, depression) has been reported during combined treatment with CYP2D6 inhibitors (e.g. quinidine, fluoxetine, paroxetine) and timolol.

Recommendations based on specific guidelines

EMA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Tiotropium

Spiolto Respimat, Spiriva, Spiriva Respimat, Tiospirex, Tiotropium Bromide

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In vitro experiments with human liver microsomes and human hepatocytes suggest that a fraction of the administered dose (74% of an intravenous dose is excreted unchanged in the urine, leaving 25% for metabolism) is metabolized by cytochrome P450-dependent oxidation and subsequent glutathione conjugation to a variety of Phase II metabolites.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Tolbutamide || CYP2C9

Tobumide, Tolbutamide, Tolmide

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Tolperisone

Tolperisone

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Tolterodine

Detrusitol, Tolterodine

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Tramadol || CYP2D6

Acugesic, Centrasic, Dolgesik, Dolocap, Durotram, Forgesic, Mabron, Orasic, Pengesic, Radol, Sefmal, Stronginal, Thramed, Tracidol, Tradol, Tradonal, Tradosik, Tradyl, Tramadol, Tramadol Stada, Tramal, Tramofal, Tugesal, Zephanal

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Use tramadol label recommended age- or weight-specific dosing.
Reduced O-desmethyltramadol (active metabolite) formation.

Recommendations based on specific guidelines

CPIC	**Use tramadol label recommended age- or weight-specific dosing. If no response and opioid use is warranted, consider a non-codeine opioid.**
DPWG	1. Be alert to a reduced effectiveness. 2. In the case of inadequate effectiveness: try a dose increase. If this does not work: choose an alternative. Do not select codeine, as this is also metabolized by CYP2D6. Morphine is not metabolized by CYP2D6. Oxycodone is metabolized by CYP2D6 to a limited extent, but this does not result in differences in analgesia in patients. 3. If no alternative is selected: advise the patient to report inadequate analgesia. It is not possible to provide a recommendation for dose adjustment, because the total analgesic effect changes when the ratio between the mother compound and the active metabolite changes.

Caveat

Like all diagnostic tests, CYP2D6 genotype is one of multiple pieces of information that clinicians should consider in guiding their therapeutic choice for each patient. Furthermore, there are several other factors that cause potential uncertainty in the genotyping results and phenotype predictions.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

32488232   25948472

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Trimipramine

Trimipramine

START MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider alternative drug not metabolized by CYP2C19.

CYP2D6 IM: Reduced metabolism of TCAs to less active compounds. Higher plasma concentrations of active drug may increase probability of side effects. CYP2C19 UM: Increased metabolism of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Consider alternative drug not metabolized by CYP2C19. If tricyclic antidepressants (TCAs) is warranted, utilize therapeutic drug monitoring to guide dose adjustment.**

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Trimipramine || CYP2C19

Trimipramine

START

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19.

Increased metabolism of tertiary amines compared to normal metabolizers. Greater conversion of tertiary amines to secondary amines may affect response or side effects.

Recommendations based on specific guidelines

CPIC	**Avoid tertiary amine use due to potential for sub-optimal response. Consider alternative drug not metabolized by CYP2C19. Tricyclic antidepressants (TCAs) without major CYP2C19 metabolism include the secondary amines nortriptyline and desipramine. If a tertiary amine is warranted, utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Initiation
This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Trimipramine || CYP2D6

Trimipramine

START

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Consider 25% reduction of recommended starting dose.

Reduced metabolism of TCAs to less active compounds when compared to normal metabolizers. Higher plasma concentrations of active drug will increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	**Consider 25% reduction of recommended starting dose. Patients may receive an initial low dose of a tricyclic, which is then increase over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose. Utilize therapeutic drug monitoring to guide dose adjustments. Titrate dose to observed clinical response with symptom improvement and minimal (if any) side effects.**
FDA	No actionable recommendation available for this drug-gene pair.

Caveat

The application of genotype-based dosing is most appropriate when initiating therapy with a tricyclic. Obtaining a pharmacogenetic test after months of drug therapy may be less helpful in some instances, as the drug dose may have already been adjusted based on plasma concentrations, response, or side effects. Similar to all diagnostic tests, genetic tests are one of several pieces of clinical information that should be considered before initiating drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose

Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Initiation

This drug-gene interaction recommendation is useful for initiation of this drug in naive patients. For patients who are already on stable dose or have used this drug before, effective drug monitoring and clinical judgement is more important.

Tropisetron

Setrovel, Tropisetron

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Insufficient evidence showing clinical impact based on CYP2D6 genotype. Initiate therapy with recommended starting dose.

Very limited data available for CYP2D6 intermediate metabolizers.

Recommendations based on specific guidelines

CPIC	**Insufficient evidence showing clinical impact based on CYP2D6 genotype. Initiate therapy with recommended starting dose.**

Caveat

Rare CYP2D6 variants may not be included in the genotype test used and patients with rare variants may be assigned a “wildtype” (CYP2D6*1) genotype by default. Thus, an assigned “wildtype” allele could potentially harbor a no or decreased function variant. Furthermore, it is important that the genetic testing platform include testing for gene copy number to identify CYP2D6 UMs. Caution should be used regarding molecular diagnostics of CYP2D6 gene copy-number variation because commercially available genotyping results may differ between diagnostic laboratories depending on assay design. Like all diagnostic tests, CYP2D6 genotype is one of multiple pieces of information that clinicians should consider when making their therapeutic choice.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Umeclidinium

Incruse Ellipta

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

In vitro metabolism of umeclidinium is mediated primarily by CYP2D6. However, no clinically meaningful difference in systemic exposure to umeclidinium (500 mcg) (8 times the approved dose) was observed following repeat daily inhaled dosing to normal (ultrarapid, extensive, and intermediate metabolizers) and CYP2D6 poor metabolizer subjects.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Upadacitinib

Rinvoq

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

Upadacitinib is metabolized in vitro by CYP3A4 with a minor contribution from CYP2D6. CYP2D6 metabolic phenotype had no effect on upadacitinib pharmacokinetics (based on population pharmacokinetic analyses), indicating that inhibitors of CYP2D6 have no clinically relevant effect on upadacitinib exposures.

Recommendations based on specific guidelines

FDA	**No actionable recommendation available for this drug-gene pair.**

More information about this drug-phenotype can also be found at **EMA** and **HCSC**.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Valbenazine

Remleas

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Venlafaxine

Avenfax, Deprevix, Efexor, Venladex Xr, Venlafaxine, Venlex Xr, Viepax

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

No action recommended based on genotype because of minimal evidence regarding the impact on efficacy or side effects.

Decreased metabolism of venlafaxine to active metabolite O-desmethylvenlafaxine (desvenlafaxine) and decreased O-desmethylvenlafaxine : venlafaxine ratio as compared to CYP2D6 normal metabolizers.

Recommendations based on specific guidelines

CPIC	**No action recommended based on genotype because of minimal evidence regarding the impact on efficacy or side effects.**
DPWG	Avoid venlafaxine. Antidepressants that are not metabolized by CYP2D6 - or to a lesser extent - include, for example, duloxetine, mirtazapine, citalopram and sertraline. If it is not possible to avoid venlafaxine and side effects occur: reduce the dose, monitor the effect and side effects or check the plasma concentrations of venlafaxine and O-desmethylvenlafaxine. It is not known whether it is possible to reduce the dose to such an extent that the side effects disappear, while the effectiveness is maintained. In general, it is assumed that the effectiveness is determined by the sum of the plasma concentrations of venlafaxine and O-desmethylvenlafaxine. However, the side effects do not appear to be related to this sum. It is not possible to offer adequately substantiated advice for dose reduction based on the literature.

Caveat

Patients on stable and effective antidepressant medication doses without significant tolerability concerns may not benefit from dose modifications based on CYP2D6, CYP2C19, and/or CYP2B6 genotype results. Pharmacogenetic test results are one of many pieces of clinical information to be considered when optimizing antidepressant drug therapy.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

30485867

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Vernakalant

Brinavess

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Voriconazole || CYP2C19

Vfend, Vorica, Voriconazole, Voriconazole Sandoz, Vorigen, Voritrop

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2C19	*17/*17	CYP2C19 Ultrarapid Metabolizer

Recommendation

Choose an alternative agent that is not dependent on CYP2C19 metabolism as primary therapy in lieu of voriconazole.

In patients for whom an ultrarapid metabolizer genotype is identified, the probability of attainment of therapeutic voriconazole concentrations is small with standard dosing.

Recommendations based on specific guidelines

CPIC	**Choose an alternative agent that is not dependent on CYP2C19 metabolism as primary therapy in lieu of voriconazole. Such agents include isavuconazole, liposomal amphotericin B, and posaconazole. Further dose adjustments or selection of alternative therapy may be necessary due to other clinical factors, such as drug interactions, hepatic function, renal function, species, site of infection, TDM, and comorbidities.** **Recommendations based upon data extrapolated from patients with CYP2C19*1/*17 genotype**
DPWG	Use an initial dose that is 1.5x higher and monitor the plasma concentration.
PRO	Consider antifungal agents other than voriconazole if patient is a known CYP2C19 ultrarapid metabolizer.
EMA	No actionable recommendation available for this drug-gene pair.

More information about this drug-phenotype can also be found at **PMDA**.

Caveat

CYP2C19 genotyping cannot replace TDM, as other factors (i.e., drug interactions, hepatic function, renal function, species, site of infection, and comorbidities) also influence the use of voriconazole. Rare CYP2C19 variants are typically not included in common genotyping tests and patients are therefore assigned the “wild-type” (CYP2C19*1) allele by default. Thus, in rare cases, an assigned “wild-type” allele may harbor a no, decreased, or increase function variant. An individual’s predicted CYP2C19 metabolizer status may also depend on other factors, including epigenetic phenomena, diet, comorbidities, or comedications.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

33896064 25239277 30653146

Change Prescription
There is a high chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Voriconazole || CYP2C9

Vfend, Vorica, Voriconazole, Voriconazole Sandoz, Vorigen, Voritrop

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal Metabolizer

Recommendation

No actionable recommendation available for this drug-gene pair.

The EMA European Public Assessment Report for voriconazole contains warning information regarding the co-administration of drugs that are substrates, inhibitors or activators of CYP3A4, CYP2C9 or CYP2C19 due to drug-drug interactions.

Recommendations based on specific guidelines

EMA	**No actionable recommendation available for this drug-gene pair.**

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Vortioxetine

Brintellix, Vortioxetine

MODERATE RECOMMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Initiate therapy with recommended starting dose.

Reduced metabolism of vortioxetine to less active compounds when compared to CYP2D6 normal metabolizers. Higher plasma concentrations may increase the probability of side effects.

Recommendations based on specific guidelines

CPIC	Initiate therapy with recommended starting dose.

Caveat

Patients on stable and effective antidepressant medication doses without significant tolerability concerns may not benefit from dose modifications based on CYP2D6, CYP2C19, and/or CYP2B6 genotype results. Pharmacogenetic test results are one of many pieces of clinical information to be considered when optimizing antidepressant drug therapy.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Follow Standard Dosing
There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Indication
This drug-gene interaction recommendation is based on a specific diagnosis ONLY. Exercise prudent clinical judgement when using the drug for other indications.

Warfarin

Apo-Warfarin, Marevan, Notisil, Orfarin, Rheoxen, Simarc, Warfarin

INFORMATION AVAILABLE

GENE	GENOTYPE	PHENOTYPE
CYP2C9	*1/*1	CYP2C9 Normal metabolizer

Recommendation

Follow Standard Dosing Guideline

No known increase risk of adverse drug reaction or therapeutic inefficacy.

Source

The curated publications relevant to the drug-gene pair and reported ethnicity are listed below:

29704269 31259883 30688796

Follow Standard Dosing

There is a low chance of sub-optimal response or adverse reaction due to patient's genetic polymorphism, no change is necessary.

Zuclopenthixol

Clopixol, Zuclopenthixol

STRONG RECOMENDATION

GENE	GENOTYPE	PHENOTYPE
CYP2D6	*4/*9	CYP2D6 Intermediate Metabolizer

Recommendation

Use 75% of the standard dose.

The risk of side effects may be elevated. The genetic variation leads to decreased conversion of zuclopenthixol, which causes the plasma concentration to be approximately 1.35-fold higher.

Recommendations based on specific guidelines

DPWG	Use 75% of the standard dose.

Source

Publications related to local relevance for this drug-gene pair have not been published yet.

Decrease Starting Dose

Patient has altered metabolism rate or enhanced activation rate for drug indicated. Decreasing starting dose has shown to help prevent patient from experiencing adverse drug reaction.

Report Verification Receipt

This page shows the verification history for report with following information

Order ID
#8349-867687

Test Order
RxReady

History

Step	Version	Date And Time	Executed By
Report Generation	v.1	21 January 2025 (15:40)	FAD Lab Director Test
Report Verification	v.1	21 January 2025 (15:40)	FAD Lab Director Test